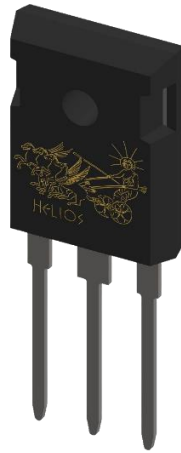


Transistor Design Project

Final Design Report

High-Speed, High-Power MOSFET



ENPH 336 Solid State Devices

Group 21

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1. Design Overview

The target specification requires the device to block 600 V when off, while conducting a continuous drain current $I_{D,cont} = 50$ A when turned on. These two requirements pull the design in opposite directions and a conventional planar NMOS cannot satisfy both simultaneously. The lateral drift region of an LDMOS is the minimal modification that resolves the conflict, and it is thus adopted.

In a conventional planar NMOS, the drain sits directly against the channel. The p-type body doping therefore controls two things at once, through the depletion width that the drain-body junction can sustain: the threshold voltage and the off-state breakdown voltage. A threshold voltage near 2 V demands a body doping on the order of 10^{16} to 10^{17} cm⁻³, whereas a breakdown voltage near 600 V requires a doping one to two decades lower. One doping region cannot do both.

The LDMOS breaks this constraint by inserting a lightly doped n-type drift region between the channel and drain contact. The channel now sits in a p-body, doped solely for threshold control, while the drift region is doped and dimensioned solely for breakdown. This decouples the specifications with two independent regions.

The drift region also distributes the electric field in the off-state. In a planar device, this peaks sharply at the drain-body junction and triggers avalanche significantly below the rated voltage target of 600 V. In the LDMOS the depletion region is extended laterally through the drift length, deconcentrating the field. This allows the breakdown to be limited by material instead of geometry. The same architecture benefits the on-state. The current path is through a short channel in series with a longer but wide-bandgap drift region, which contributes little resistance due to the large W/L ratio. The drift region dominates contribution of R_{on} and its resistance is kept manageable by the high critical field of GaN, which permits a short drift length for a given breakdown voltage. The choice of LDMOS and the choice of GaN therefore makes a good pairing.

The LDMOS is the minimum topology in which the specifications can be met. All subsequent sections are organized around the two-region structure established here.

1.1 Device Type: NMOS vs. PMOS

For a high-speed, high-power MOSFET, the team choose to design an n-channel (NMOS) device.

The key consideration between designing an NMOS or PMOS device was the choice of the primary charge carrier and its mobility. In NMOS devices, electrons are the primary charge

carriers, and in PMOS devices, holes are the primary charge carriers. In gallium nitride (GaN), which is the semiconductor substrate the team used, electrons have a higher mobility ($1500 - 2000 \text{ cm}^2/\text{Vs}$) compared to holes ($\sim 200 - 400 \text{ cm}^2/\text{Vs}$). This translates to electrons moving faster, allowing an NMOS device to switch states faster than a PMOS device, which is vital for a high-speed design. Additionally, since $R_{on} \propto \frac{1}{\mu_{p,n}}$, an NMOS device's larger mobility presents a lower resistance.

Furthermore, to deliver the same performance, PMOS devices require a larger surface area. The overall larger size of a PMOS device that performs the same as an NMOS counterpart presents numerous speed limitations. For example, a larger gate area would increase the gate capacitance (C_g), which would consequently mean that it would take longer for the gate to charge and discharge, effectively hindering high-speed performance. NMOS's smaller design helps maintain high switching speeds.

1.2 Substrate Material Selection

As mentioned above, the semiconductor substrate the team used was gallium nitride (GaN) due to its fast-switching speed, efficiency at high switching frequency, small switching losses, and power density.

The selection of GaN was made after the consideration of three substrate materials: Silicon, GaN, and Silicon Carbide (4H-SiC). The following discussion provides justification for choosing GaN over silicon and 4H-SiC, which uses key material properties summarized in Table 1 below.

Table 1: Intrinsic material properties of Silicon, Gallium Nitride, and the 4H polytype of Silicon Carbide

Physical Property	Silicon (Si)	Gallium Nitride (GaN)	Silicon Carbide (4H-SiC)
Bandgap Energy (eV)	1.12	3.40	3.26
Critical Breakdown Field (MV/cm)	0.3	3.3	2.8
Electron Mobility ($\text{cm}^2/\text{V}\cdot\text{s}$)	1400	1500 - 2000	650 - 900
Thermal Conductivity ($\text{W}/\text{cm}\cdot\text{K}$)	1.5	1.3	4.9
Maximum Junction Temperature ($^{\circ}\text{C}$)	150	>150 - 200	> 200

Intrinsic Carrier Concentration (cm⁻³)	1.0*10 ¹⁰	1.9*10 ⁻¹⁰	8.2*10 ⁻⁹
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When compared to silicon, GaN, which is a wide bandgap (WBG) material, offers a wide bandgap and a significantly higher critical electric field. Since the breakdown voltage is proportional to the critical electric field, $V_B \propto E_{crit}^2$ and since the breakdown field of GaN is approximately ten times as large as that of silicon, this means that for the same performance, a GaN device can be made to be approximately ten times thinner compared to a silicon device. Furthermore, electron mobility is larger in GaN compared to silicon, enabling GaN to switch at higher frequencies. Lastly, GaN offers a higher maximum junction temperature, which is necessary for high power applications.

When compared to 4H-SiC, which is another WBG material, the biggest advantage of GaN is its significantly larger (double) electron mobility. However, 4H-SiC excels in its significantly higher thermal conductivity, which indicates that it can pull heat away from the junction three times more effectively. While this is an advantage, the reduced electron mobility of 4H-SiC profoundly hinders its ability to support a high-speed device.

1.3 Dopant Selection

The MOSFET has four regions, each with a different doping role. Source and drain are heavily doped n-type contact regions providing low contact resistance. Drift is a lightly doped n-type layer between channel and drain that supports the breakdown voltage. The body is a bulk p-type substrate in which the MOS capacitor is formed; the doping here sets threshold voltage.

The team's initial plan for the p-doped body was to use magnesium (Mg) as the acceptor. While in silicon doped with boron at 25°C, the assumption of 100% acceptor ionization is reasonable, this is not true for gallium-nitride doped with magnesium (Semiconductor Physics and Devices, Donald A. Neamen). This can be shown through evaluating the equation below:

$$\frac{p_a}{p_0 + p_a} = \frac{1}{1 + \frac{N_v}{4N_a} \cdot \exp\left(\frac{-(E_a - E_v)}{k_B T}\right)}$$

Silicon with boron ([1]): $\frac{p_a}{p_0 + p_a} = 0.04$ 0.96 N_a ionized at 300K	GaN with magnesium ([2]): $\frac{p_a}{p_0 + p_a} = 0.84$ 0.16 N_a ionized at 300K	GaN with beryllium ([3]): $\frac{p_a}{p_0 + p_a} = 0.39$ 0.61 N_a ionized at 300K
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*Note: The material properties N_v and $(E_a - E_v)$ were found online. N_a was set to $8 * 10^{16}$.*

This table shows how Be doped GaN would experience higher ionization at room temperature than Mg doped GaN. This fact was behind the team's decision to use Be. While Be provides better ionization, it is toxic to work with and can cause compensation issues from unintentional donors during the doping process. The team is justifying ignoring compensation issues as the paper "Passivation of beryllium acceptor in GaN as a possible route for p-type doping" outlines a potential method to circumvent compensation issues using hydrogen bound with beryllium to dope the GaN later removing the hydrogen through annealing producing p-type GaN. [3]

The team used silicon as the donor in GaN because at 300K, using the donor version of the formula above, 100% donor ionization can be assumed.

The differential doping concentrations across these regions were determined by modeling the MOSFET behavior in MATLAB and tuning parameters until the output matched the team's desired results. This mathematical process is described in Performance Calculations. The team compared two Infineon transistors, a silicon power MOSFET [4] and a GaN HEMFET [5], to determine if the mathematical model outputs were desirable. This comparison was made to ensure the team's final design was semi-competitive in the power transistor industry.

1.4 Gate Oxide and Contacts

Silicon dioxide, SiO_2 , was used for the gate oxide since it serves as an excellent insulator that enables high input impedance, which enables the gate to control the channel while preventing DC current flow from the gate. It also has a high critical field allowing for increased gate voltages without material failure.

Aluminium was chosen as the gate metal as it has low resistivity and is the most common gate metal used in MOSFET fabrication.

Ohmic metal-semiconductor contacts were used for the source and drain contacts since they serve as a low-resistance junction allowing conduction in both directions. This is vital for high-speed and high-power performance as high resistance or rectifying behaviour would degrade device performance.

The Ohmic contacts are tunneling barriers, opposed to an ideal nonrectifying barrier. This is because finding a metal with a work function Φ_m less than the work function of GaN Φ_s is difficult and the wide bandgap of 3.40 eV for GaN makes it not possible to create low enough barriers for an ideal rectifying junction.

The metal used for the source and drain contacts was a stack of Ti/Al/Ni/Au. This stack is required as the metal interfacing with the GaN, titanium, oxides in air degrading device performance significantly. Once this stack is formed on top of the GaN it is annealed to improve the quality of the interface, improving the contacts performance significantly. This method is outlined by the research article “Low Resistivity n-type GaN Ohmic Contacts on GaN Substrates”. [6]

1.5 Device Geometry

The final device design had a geometry as seen in Figure 1, following the parameters in Table 2 below. The height of each component in the Ti/Al/Ni/Au source-drain contact stack was determined following the research article’s design.

Table 2: Final device geometry considered in calculations

Parameter	Description	Value
W	Device Width	$50,000 \mu m$
H	Device Height	$5 \mu m$
L	Device Length	$19 \mu m$
H_{n+}	Source/Drain Region Height	$0.2 \mu m$
H_g	Gate Height	$1 \mu m$
H_c	Contact Height	$0.740 \mu m$
	$H_{C,Ti}$ Contact titanium height	$20 nm$
	$H_{C,Al}$ Contact aluminum height	$300 nm$
	$H_{C,Ni}$ Contact nickel height	$20 nm$
	$H_{C,Au}$ Contact gold height	$400 nm$
t_{ox}	Oxide Thickness	$30 \mu m$
$L_{ov,B}$	Source/Body Contact overlap	$2 \mu m$
L_c	Source/Drain Contact Length	$4 \mu m$
L_g	Gate Length	$1.4 \mu m$
L_{ch}	Channel Length	$1 \mu m$
L_{ov}	Source/Drift Gate Overlap	$0.2 \mu m$
L_{drift}	Drift Length	$6 \mu m$
L_{n+}	Source/Drain Region Length	$5 \mu m$

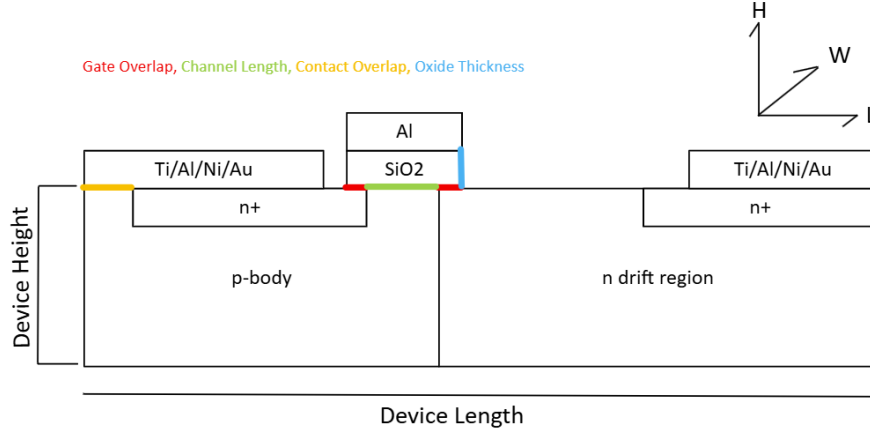


Figure 1: Slice illustrating the geometry of the final design. Not to scale.

Because of the design requirement for fast switching, a continuous 50mm long MOSFET would not be viable due to material property issues of the geometry being centimeters long and only micrometers wide and tall. This would also result in massive gate resistance which limits switching speeds. This issue is solved in the industry with multi-finger MOSFET transistor designs [7]. The idea is to split the transistor into thinner “fingers” and use interconnected metal fingers to wire all the contacts together in parallel. This lowers gate resistance, allowing for faster switching speeds. The teams design features 50 fingers, each 1mm wide. This design choice allows for gate resistance of $125m\Omega$. The calculation for gate resistance can be seen below.

$$R_g = \frac{1}{3} \frac{R_{sheet} * W}{N^2 * L} = \frac{1}{3} \frac{\rho * W}{N^2 * L * H} = \frac{1}{3} * \frac{2.826 * 10^{-8} * 50 * 10^{-3}}{50^2 * 1.4 * 10^{-6} * 1 * 10^{-6}} = \mathbf{135\ m\Omega}$$

Note: The values for aluminum rho are from an online source. [8] The $\frac{1}{3}$ factor comes from the behavior of all the 1mm slices in parallel with single-sided connections. [9]

This value is significantly lower than the gate resistances featured on the datasheets the team used as reference points during the design process. This is because the above calculation is only for the resistance of the Al gate. It does not account for resistances from package the transistor is in or any other nonidealities.

Following the Neaman equations, minimum drift thickness is calculated below.

$$L_{drift,min} = \frac{2V_B}{E_{crit}} = \frac{2 * 600}{3 * 10^8} = 4\ \mu m \rightarrow L_{drift} = 1.5L_{drift,min} = \mathbf{6\ \mu m}$$

The final design features a multiplier of 1.5 to this minimum drift thickness as it ensures a safety margin to the critical field.

1.6 Design Methodology

The team's design methodology was to find and combine equations required to create a mathematical model of the MOSFET design and isolate mathematically independent parameters which could be tuned. The model was simulated in MATLAB and then each parameter was tweaked until the output was deemed desirable (competitive with similar transistors in the industry). The MATLAB model can be found in Appendix B: MATLAB Calculations, and the final output can be found in Appendix A: MATLAB Calculations Output.

The independent parameters which could be individually tuned to adjust the models output were the following: Gate drive (V_G), Breakdown voltage (V_b), Gate width (W), channel length (L), oxide thickness (t_{ox}), gate overlap length (L_{ov}), contact length (c_l), device height (H), donor doping level (N_d), and acceptor doping level (N_a).

2. Performance Calculations

Note that all calculations below occur at $T = 300\text{ K}$ (25°C), which is the typical operating temperature, unless otherwise specified.

2.1 Gate Oxide Electric Field Limit

The gate oxide thickness is defined as:

$$t_{ox} = 30\text{ nm}$$

The electric field across the diode is:

$$E_{ox} = \frac{V_g}{t_{ox}}$$

At a gate voltage of $V_g = 10\text{ V}$:

$$E_{ox} = \frac{10}{30 * 10^{-9}} = 3.33 \frac{\text{MV}}{\text{cm}}$$

This value remains safely below the breakdown field of SiO_2 which is approximately 10 MV/cm , confirming the oxide reliability under normal conditions. [10]

2.2 Acceptor Ionization

The drain/source regions are heavily doped n-type, and the channel body is lightly doped p-type using beryllium (Be) as the acceptor in GaN. The raw acceptor concentration is $N_{a,raw}$

= $1.5 \times 10^{17} \text{ cm}^{-3}$, but Be in GaN has an ionization energy of $E_{\text{Be}} = 116 \text{ meV}$ well above the thermal energy at 300 K ($\sim 25 \text{ meV}$) so only a fraction of dopants are ionized at room temperature.

Fraction of un-ionized Be acceptors (Fermi-Dirac occupancy) is given by:

$$f = \frac{1}{1 + \frac{N_v}{4N_a} \exp\left(-\frac{E_A}{kT}\right)}$$

where $N_v(T) = N_{v,300}(T/300)^{3/2}$ is the temperature-dependent effective valence-band density of states. With $N_{v,300} = 4.6 \times 10^{19} \text{ cm}^{-3}$ and $E_{\text{Be}} = 116 \text{ meV}$, approximately 67.5 % of the Be atoms remain un-ionized at 300 K, yielding an effective channel doping of

$$N_a = N_{a,\text{raw}}(1 - f) = 4.877 * 10^{16} \text{ cm}^{-3}$$

The justification for the specific choice of $N_{a,\text{raw}}$ and N_d values, including the Be doping process considerations, is provided in Dopant Selection.

2.3 Drift Region Design

The drift region is designed to sustain the full blocking voltage $V_b = 600 \text{ V}$ using the one-sided abrupt junction approximation. Two equations govern the drift background doping N_b and the required drift width W_{drift} . [11] [12]

Background doping from breakdown condition:

$$N_b = \frac{\epsilon_s E_{\text{crit}}^2}{2qV_b} = 3.68 * 10^{15} \text{ cm}^{-3}$$

Drift region width from depletion approximation:

$$W_{\text{drift}} = \frac{2V_b}{E_{\text{crit}}} = 4.0 \text{ }\mu\text{m}$$

With $E_{\text{crit}} = 3 \times 10^6 \text{ V/cm}$ (GaN critical field) and $\epsilon_s = 8.9\epsilon_0$ (GaN permittivity).

2.4 Electron Mobility

The electron mobility in the channel body is computed using the Mnatsakanov empirical model, which accounts for both temperature and impurity scattering. [13] [14]

$$\mu_n(T, N) = \mu_{\text{min}} + \frac{\mu_{\text{max}} \left(\frac{300}{T}\right)^{\beta_1} - \mu_{\text{min}}}{1 + \left(\frac{N}{N_{\text{REF}}} \left(\frac{T}{300}\right)^{\gamma}\right)^{\lambda}}$$

Channel mobility:

$$\mu_{n,ch} = \mu_n(T, N_a) = 1201.5 \frac{cm^2}{V \cdot s}$$

Drift region mobility:

$$\mu_{n,drift} = \mu_n(T, N_b) = 1347.7 \frac{cm^2}{V \cdot s}$$

Parameters: $\mu_{max} = 1600 \text{ cm}^2 / V \cdot s$, $\mu_{min} = 100 \text{ cm}^2 / V \cdot s$, $N_{ref} = 3 \times 10^{17} \text{ cm}^{-3}$, $\beta_1 = 3$, $\gamma = 4.4$, $\lambda = 0.7$.

2.5 Thermal Voltage

$$V_t = \frac{kT}{e} = 25.70 \text{ mV}$$

2.6 GaN Bandgap

$$E_g = E_g(0 \text{ K}) - \frac{\alpha T^2}{T + \beta} = 3.435 \text{ eV}$$

Parameters: $E_g(0 \text{ K}) = 3.51 \text{ eV}$, $\alpha = 9.09 \times 10^{-4} \text{ eV/K}$, $\beta = 830 \text{ K}$. [15]

2.7 Intrinsic Carrier Concentration

$$n_i(T) = \frac{n_{i,300} \left(\frac{T}{300} \right)^{\frac{3}{2}} \exp \left(\frac{-E_g(300)}{2 \cdot 0.02585} \right)}{\exp \left(\frac{E_g(T)}{2(0.02585)} \right)} = 1.9 * 10^{-4} \text{ m}^{-3}$$

2.8 Threshold Voltage

The threshold voltage V_{th} determines the gate-source voltage at which the inversion channel forms. The calculation follows the standard MOS capacitor analysis for a p-type body (GaN) with an SiO₂ gate dielectric and an aluminum gate metal. [10] [16] [17]

2.8.1 Oxide Capacitance

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.151 * 10^{-3} \frac{F}{cm^2}$$

2.8.2 Fermi Potential

The Fermi potential ϕ_{fp} represents the energy separation between the Fermi level and the intrinsic level in the p-type body

$$\phi_{fp} = V_t \ln \left(\frac{N_a}{n_i} \right) = 1.57 \text{ V}$$

2.8.3 Maximum Depletion Width and Charge

Maximum depletion width at strong inversion:

$$x_{dT} = \sqrt{\frac{4\epsilon_s \phi_{fp}}{qN_a}} = 0.31 \mu\text{m}$$

Maximum ionized depletion charge per unit area:

$$Q_{SD,max} = qN_a x_{dT} = 1.968 * 10^{-3} \frac{\text{C}}{\text{m}^2}$$

2.8.4 Metal-Semiconductor Work Function Difference

An aluminum gate metal is used ($\phi_m = 4.38 \text{ eV}$). The GaN electron affinity is $\chi = 4.1 \text{ eV}$ [18].

The work function difference is:

$$\phi_{ms} = \phi_m - \left(\chi_{GaN} + \frac{E_g}{2} + \phi_{fp} \right) = -3.011 \text{ V}$$

2.8.5 Threshold Voltage

$$Q'_{ss} = 10^{11} \text{ cm}^{-2}$$

Combining all contributions, the threshold voltage is:

$$V_{th} = \frac{Q_{SD,max}}{C_{ox}} - \frac{qQ'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp} = 1.7066 \text{ V}$$

The minimum and maximum values of V_{th} were determined by shifting the parameters t_{ox} , N_a , and Q_{ss} . For the minimum V_{th} value $0.9t_{ox}$, $0.9N_a$, and $1.2Q_{ss}$ were used. For the maximum, $1.1t_{ox}$, $1.1N_a$, and $0.8Q_{ss}$ were used. These values were arbitrarily determined by the team to simulate imperfect manufacturing.

2.9 On-State Resistance

The On-State resistance is the combination of the channel resistance, the drift resistance, and the source/drain contact resistances.

$$R_{DS(ON)} = R_{source \text{ contact}} + R_{drain \text{ contact}} + R_{drift} + R_{ch}$$

The channel resistance was calculated using the following Neaman formula.

$$R_{ch} = \frac{L}{W} \cdot \frac{1}{\mu_{n,ch} C_{ox} (V_{GS} - V_{th})} = 17.437474553929 \text{ m}\Omega$$

The drift resistance was calculated using the following Neaman formulas:

$$R = \frac{L}{\sigma A}, \quad \sigma \cong e\mu_n N_d \rightarrow R_{drift} = \frac{L}{e\mu_{n,drift} N_d A}$$

$$R_{drift} = \frac{L}{e\mu_{n,drift} N_d A} = 30.132398197220 \text{ m}\Omega$$

The process to calculate the source/drain contact resistances required formulas that are not present in Neaman's textbook. While the ohmic contacts in the design are ideally only tunneling, the effect of the phonons needs to be accounted for. The equation in the research article "*Low Resistivity n-type GaN Ohmic Contacts on GaN Substrates*" [6] calculates the contact resistivity with the TFE model, which is a combination of the TE and FE formulas present in Neaman's textbook. The condition for this equation to be valid is that $\frac{k_B T}{E_{00}} \cong 1$.

$$\phi_B = \phi_m - \chi_s = 4.33 - 4.1 = 0.23 \text{ eV}$$

$$E_{00} = \frac{e\hbar}{2} \sqrt{\frac{N_d}{\epsilon_s m_n^*}} = 7.049849106138426 * 10^{-21} \text{ J} = 0.0440016972 \text{ eV}$$

$$A^* = \frac{4\pi e m_n^* k_B^2}{h^3} = 24.04498051661899 \text{ A}/(\text{cm}^2 * \text{K}^2)$$

$$\phi_n = \frac{k_B T}{e} \ln\left(\frac{N_c}{N_d}\right) = -0.038012009188394 \text{ eV}$$

Because $\frac{k_B T}{E_{00}} = 0.59$, we can use the TFE model to estimate contact resistivity.

$$\rho_c = \frac{k^2}{qA^*} \frac{1}{\sqrt{\pi(\phi_B + \phi_n)E_{00}}} \cosh\left(\frac{E_{00}}{k_B T}\right) \sqrt{\coth\left(\frac{E_{00}}{k_B T}\right)} \exp\left(\frac{\phi_B + \phi_n}{E_{00} \coth\left(\frac{E_{00}}{k_B T}\right)} - \frac{\phi_n}{k_B T}\right)$$

$\frac{k^2}{qA^*} = 4.951085382592316 * 10^{-29}$	$\frac{1}{\sqrt{\pi(\phi_B + \phi_n)E_{00}}} = 3.8314826667 * 10^{19}$
$\cosh\left(\frac{E_{00}}{k_B T}\right) = 2.860770025547636$	$\sqrt{\coth\left(\frac{E_{00}}{k_B T}\right)} = 1.033117729891008$
$\exp\left(\frac{\phi_B + \phi_n}{E_{00} \coth\left(\frac{E_{00}}{k_B T}\right)} - \frac{\phi_n}{k_B T}\right) = 261.5037366606163$	

$$\rho_c = 1.466148436247367 * 10^{-6} \Omega \text{ cm}^2$$

$$R_{source\ contact} = R_{drain\ contact} = \frac{\rho_c}{A} = 0.7330742181236835\ m\Omega$$

$$R_{DS(on)} = R_{source\ contact} + R_{drain\ contact} + R_{drift} + R_{ch} = \mathbf{49\ m\Omega}$$

To achieve the minimum and maximum $R_{DS(on)}$ values present on the pamphlet, the team simulated the minimum resistance by shifting N_d, N_a by +10%, L by -5%, and t_{ox} by -10% and the maximum resistance by shifting N_d, N_a by -10%, L by +5%, and t_{ox} by +10%. These values were arbitrarily selected by the team to simulate manufacturing defects. This was done to simulate how transistor manufacturing in the industry is not a perfect process.

2.10 Maximum Continuous & Pulse Drain Current

2.10.1 Maximum Continuous Drain Current

Although the device geometry can handle the above large saturation current, the drain-to-source current that the device can safely operate at is determined by the device's package thermal limits. To determine the thermal current limits of the power MOSFET and hence determine the maximum continuous drain-to-source current, the following calculations are performed. All calculations and formulas below are based on guidelines published in Texas Instruments (TI) Application Report SLUAAT8 [19].

Initially, key thermal parameters are defined as follows, beginning with the maximum junction temperature, $T_{j,max}$, which is set to 125°C. Based on Infineon power MOSFETs which also use the TO-247 package, a conservative junction-to-case thermal resistance, $R_{th,JC} = 0.5\ ^\circ\text{C}/W$ is used. Additionally, since this high-power package features a large, exposed metal tab on the back that allows for direct thermal coupling to a heat sink, case temperature values are kept close to ambient temperatures. Hence, a $T_c = 25^\circ\text{C}$ is chosen. Lastly, the temperature coefficient, $\alpha = 0.005\ \frac{1}{^\circ\text{C}}$ is chosen as a standard linear approximation for the drain-source on resistance, $R_{DS(on)}$.

To calculate the $R_{DS(on)}$ at $T_{j,max}$, the R_{on} is normalized as per TI recommendations:

$$Normalization\ Factor = 1 + \alpha(T_{j,max} - 25) = 1.5$$

$$R_{DS(on)@T_{j,max}} = R_{on} * Normalization\ Factor = 49m\Omega * 1.5 = 73.5\ m\Omega$$

The power dissipation limit is defined by the thermal resistance between the junction and the case. Maximum power dissipation is calculated using TI equation 5.

$$P_{max} = \frac{T_{j,max} - T_c}{R_{th,JC}} = \frac{125 - 25}{0.5} = 200W$$

Therefore, under the assumption of a constant case temperature, which assumes an efficient heat sink, the maximum power dissipation for the MOSFET is set to 200 W. Now, the maximum continuous drain current, $I_{D,max}$, is calculated using TI equation 3:

$$I_{D,max} = \sqrt{\frac{P_{max}}{R_{DS(on)}@T_{j,max}}} = \sqrt{\frac{200}{73.5m}} = 52.14 \text{ A}$$

2.10.2 Maximum Pulsed Drain Current

For short-duration pulses, such as 100 μ s, the MOSFET is able to handle a higher current compared to $I_{D,max}$. To model and calculate this, transient thermal impedance, $Z_{\theta JC}$, is used. $Z_{\theta JC}$ is approximated using the one-dimensional heat diffusion approximation, such that $Z_{\theta JC} = \sqrt{\frac{t_p}{t_{ref}}}$, where $t_p = 100\mu s$ is the test pulse and the reference is 1.0 s.

To calculate the maximum pulsed drain current, TI equation 4 is used below:

$$I_{D,max,pulsed} = \sqrt{\frac{T_{j,max} - T_c}{Z_{\theta JC} R_{th,JC} R_{DS(on)}@T_{j,max}}} = 521.45 \text{ A}$$

As seen above, with a 100 μ s pulse, the theoretical maximum pulsed drain current is 521.45 A. However, as calculation below, the maximum current the device can handle in saturation is 237 A, thus the maximum pulsed drain current gets set to 237 A. This is roughly four times larger than the maximum continuous drain current, which meets industry standards.

2.11 Gate-Source Plateau Voltage

During the turn-on switching transient, V_{GS} passes through a constant-voltage region known as the Miller plateau. At this voltage the gate current is entirely consumed charging C_{gd} (the Miller capacitor) while V_{DS} falls from the bus voltage to the on-state value. Knowing $V_{GS,pl}$ is essential for predicting dV_{DS}/dt , switching loss, and dV/dt -induced false turn-on immunity. [20] [21]

2.11.1 Formula Derivation

In saturation, the drain current is:

$$I_D = \left(\frac{\mu_n C_{ox} W}{2L} \right) (V_{GS} - V_{th})^2 = 237 \text{ A}$$

At the plateau, I_D equals the load current $I_{D,max}$ so V_{GS} is pinned. Using the thermally limited continuous drain current $I_{D,max} = 52.14$ A, rather than the value of 237 A obtained from saturation, and the device parameters at $T = 300$ K:

$$V_{gs,pl} = V_{th} + \sqrt{\frac{2I_{D,max}}{\mu_n C_{ox} \left(\frac{W}{L}\right)}} = 5.59 \text{ V}$$

The plateau lies between $V_{th} = 1.71$ V and the drive voltage $V_G = 10$ V, confirming a well-defined Miller region.

To calculate the minimum and maximum values present on the pamphlet, the same process used in the 2.8 Threshold Voltage calculations was applied.

2.12 Gate Capacitances

The intrinsic MOSFET gate capacitances vary with operating region because the charge distribution in the channel changes. Three distinct regions are modeled: saturation, triode (linear), and cutoff. These capacitances directly determine switching speed and gate charge requirements. [10] [22] [23]

2.12.1 Saturation Region ($V_{DS} \geq V_{GS} - V_{th}$)

$$C_{gs,sat} = \frac{2}{3} W L C_{ox} + W L_{ov} C_{ox} = 49.88 \text{ pF}$$

$$C_{gd,sat} = W L_{ov} C_{ox} = 11.51 \text{ pF}$$

2.12.1.1 C_{ds} Model

To model the drain to source capacitor C_{ds} , the following method was used, drawing upon a paper from [24].

$$C_{j0} = \sqrt{\frac{2\epsilon_s \left(\frac{N_a N_d}{N_a + N_d}\right)}{2\phi_0}}$$

$$A_j = W x_{dT}$$

$$\phi_0 = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

$$K_{eq} = (-2 \sqrt{\frac{\phi_0}{(V_2 - V_1)}})(\sqrt{\phi_0 - V_2} - \sqrt{\phi_0 - V_1})$$

$$C_{ds} = A_j C_{j0} K_{eq} = 0.575 \text{ pF}$$

2.12.2 Triode (linear) Region ($V_{DS} < V_{GS} - V_{th}$)

$$C_{gs,tri} = C_{gd,tri} = \frac{1}{2}WL C_{ox} + WL_{ov}C_{ox} = 40.29 \text{ pF}$$

2.12.3 Cutoff Region ($V_{GS} \leq V_{th}$)

$$C_{gs,cut} = C_{gd,cut} = WL_{ov}C_{ox} = 11.51 \text{ pF}$$

2.12.4 Datasheet Equivalents

$$C_{iss} = C_{gs} + C_{gd} = 61.39 \text{ pF}$$

$$C_{oss} = C_{ds} + C_{gd} = 12.09 \text{ pF}$$

2.13 Switching Time Analysis & Gate Charge

To analyze and calculate the switching time and gate charge characteristics of a power MOSFET, the framework and guidelines provided in the Toshiba MOSFET Grade Drive Circuit Application Note are used below [25].

2.13.1 Switching Time Analysis

The switching characteristics of a MOSFET are divided into the specific periods based on the charging and discharging of the internal gate capacitances, C_{gd} and C_{gs} . The total input capacitance, C_{iss} , calculated above, is used. Although the intrinsic internal gate resistance is $R_{g,int} = 135 \text{ m}\Omega$, to account for additional resistance in the drive circuitry, a value of $R_g = 6 \text{ }\Omega$ is used. The gate drive used is $V_G = 10 \text{ V}$, which is the recommended value.

2.13.1.1 Turn-On Delay

The turn-on process of a power MOSFET begins with the turn-on delay, $t_{d(on)}$, which is the time it takes for the gate voltage to rise to the threshold voltage V_{th} . Using Toshiba equation 1, this delay is calculated.

$$t_{d(on)} = R_g(C_{gs} + C_{gd}) \ln\left(\frac{V_G}{V_G - V_{th}}\right)$$

However, to account for common source inductance, L_s , in the package, which opposes the gate drive, an inductive delay component is added to the above formula, resulting in the following $t_{d(on)}$ calculation. A source inductance of $L_s = 2 \text{ nH}$ is used.

$$t_{ind} = L_s * \frac{I_{D,max}}{V_G - V_{th}}$$

$$t_{d(on)} = R_g(C_{gs} + C_{gd}) \ln\left(\frac{V_G}{V_G - V_{th}}\right) + t_{ind} = 12.6440 \text{ ns}$$

2.13.1.2 Rise Time

The overall rise time, t_r , begins when the gate voltage reaches the threshold voltage and ends when it reaches the Miller plateau voltage, $V_{GS(pl)}$. During this time, C_{gs} and C_{gd} are charged in parallel, and as the gate voltage rises, the drain current starts to flow and increases. This corresponds to a current rise time, t_{ir} , which is calculated using Toshiba equation 2.

$$t_{ir} = R_g(C_{gs} + C_{gd}) \ln \left(\frac{V_G - V_{th}}{V_G - V_{GS(pl)}} \right)$$

During period $t_3 - t_2$ in Toshiba, the gate voltage is constant at the Miller plateau, while V_{DS} decreases. This corresponds to a voltage fall time, t_{vf} , which is calculated using Toshiba equation 3.

$$t_{vf} = \frac{Q_{gd} R_g}{V_G - V_{GS(pl)}} = R_g C_{gd} \frac{V_{DS}}{V_G - V_{GS(pl)}}$$

The overall rise time is the combination of these two:

$$t_r = t_{ir} + t_{vf} = \mathbf{8.0630 \text{ ns}}$$

2.13.1.3 Turn-Off Delay

The turn-off delay, $t_{d(off)}$, which is the time needed to discharge the gate from the full gate drive voltage V_G to $V_{GS(pl)}$, represents the time when the MOSFET is on even though the drive signal has been removed. $t_{d(off)}$ is calculated using the following formula, with the same added effect of the source inductance as discussed above.

$$t_{d(off)} = R_g(C_{gs} + C_{gd}) \ln \left(\frac{V_G}{V_{GS(pl)}} \right) + t_{ind} = \mathbf{12.7893 \text{ ns}}$$

2.13.1.4 Fall Time

The overall fall time, t_f , is calculated based on the voltage rise time, t_{vr} , and the current fall time, t_{if} . When the gate voltage reaches the Miller plateau, V_{DS} begins to rise towards the supply level, corresponding to the following t_{vr} equation.

$$t_{vr} = R_g C_{gd} \frac{V_{DS}}{V_{GS(pl)}}$$

Once the drain voltage is at the supply level, the MOSFET enters the current fall phase which is calculated below.

$$t_{if} = R_g(C_{gs} + C_{gd}) \ln \left(\frac{V_{GS(pl)}}{V_{th}} \right)$$

The overall fall time is the sum of these two times:

$$t_f = t_{vr} + t_{if} = \mathbf{6.6140\ ns}$$

2.13.1.5 Switching Frequency

The maximum theoretical switching frequency, f_{sw} , is dictated by the sum of all the above transition times, as calculated below.

$$f_{sw} = \frac{1}{t_r + t_f + t_{d(on)} + t_{d(off)}} = \mathbf{24.93\ MHz}$$

It is important to note that the above switching frequency is a theoretical maximum based on the above calculated timing parameters. Since the calculations use saturation capacitances, which are smaller than real world measured values, the timing values are ideal. As such, the theoretical maximum switching frequency found using these smaller ideal timing values is expected to be much larger than real world maximums. Furthermore, as discussed and concluded in 2.14 Power Analysis & Switching Frequency below, the recommended maximum switching frequency is limited by the power consumption characteristics.

2.13.2 Gate Charge

The total gate charge, Q_g , is the sum of Q_{gs} and Q_{gd} , representing the charge that the gate drive circuit must deliver to fully turn on the MOSFET.

Q_{gs} is the charge required to bring the gate up to the Miller plateau, and is comprised of the charge to reach threshold, Q_{gs1} , and the charge to reach plateau from threshold, Q_{gs2} . These three charges are calculated below:

$$Q_{gs1} = C_{iss} * V_{th} = 0.1048\ nC$$

$$Q_{gs2} = C_{iss} * (V_{GS(pl)} - V_{th}) = 0.2384\ nC$$

$$Q_{gs} = Q_{gs1} + Q_{gs2} = \mathbf{0.3432\ nC}$$

Q_{gd} , which is the charge required to sweep the drain-source voltage from the supply down to the on-state voltage, is calculated using the voltage fall time:

$$Q_{gd} = I_{gate} * t_{vf} = \left(\frac{V_G - V_{GS(pl)}}{R_g} \right) * t_{vf} = \mathbf{5.7551\ nC}$$

Lastly, total gate charge is calculated:

$$Q_g = Q_{gs} + Q_{gd} = \mathbf{6.0983\ nC}$$

2.14 Power Analysis & Switching Frequency

The total power dissipation of a MOSFET depends on the conduction, switching, output capacitance, and gate drive power losses.

Conduction power loss occurs when the MOSFET is fully enhanced, acts as a closed switch, and has a small internal resistance. It is calculated using the equation below, where the duty cycle is set to 0.5 (50%) for a worst-case estimation.

$$P_{cond} = I_{D,max}^2 * R_{on} * D$$

Switching power loss occurs during the transitions between the on and off states, which are represented by the rise and fall times. It is calculated using the equation below:

$$P_{sw} = V_{DS} * I_{D,max} * \left(\frac{t_r + t_f}{2} \right) * f_{sw}$$

Output capacitance loss is due to the stored energy on the output capacitance that is dissipated when the MOSFET turns back on.

$$E_{coss} = \frac{1}{2} * C_{oss} * V_{DS}^2$$

$$P_{coss} = E_{coss} * f_{sw}$$

Lastly, gate drive power loss, which arises from turning on and off the device at the gate, is calculated using the following equation.

$$P_{gate} = Q_g * V_G * f_{sw}$$

Total power dissipation is the sum of all of these power losses:

$$P_{tot} = P_{cond} + P_{sw} + P_{coss} + P_{gate}$$

As derived in 2.10.1 Maximum Continuous Drain Current above, the maximum power dissipation due to thermal limits is 200 W. Since switching, output capacitance, and gate drive power loss are all proportional to f_{sw} , the thermal limited switching frequency is now determined as opposed to the theoretical maximum value obtained from timing characteristics. To do this, a MATLAB frequency sweep was performed, where for each switching frequency, the total power dissipation was calculated using $I_{D,max}$. The thermal limited switching frequency was found by finding the intersection of the graph with the defined $P_{max} = 200W$ line. Figure 2 displays this plot.

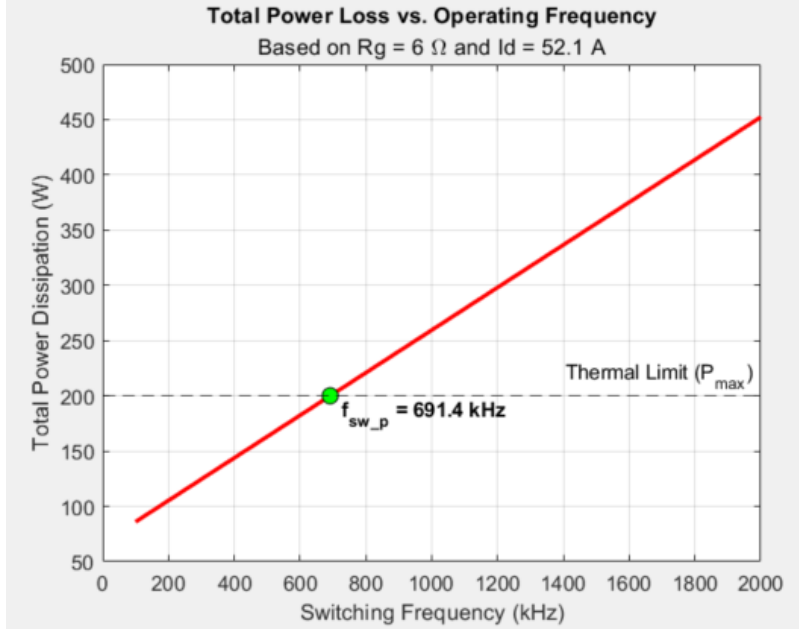


Figure 2: Power loss vs operating frequency plot.

This yielded a thermally limited $f_{sw} = 691.4 \text{ kHz}$, which offers fast switching capabilities.

With the f_{sw} known, the power dissipations were calculated, as summarized in Table 3 below.

Table 3: Power and corresponding values.

Parameter	Value (W)
P_{cond}	66.6667
P_{sw}	132.2840
P_{coss}	1.0444
P_{gate}	0.0042
P_{tot}	199.9993

Therefore, the total power dissipation at max current and thermally limited switching frequency is 200 W. If the device is operated at a lower current, the thermally limited switching frequency would increase

2.15 Body Diode Characteristics

The intrinsic body diode of the MOSFET (the p-n junction between the p-type channel body and the n-type drift/drain region) conducts during dead-time intervals. In GaN devices the body diode forward voltage is substantially higher than in silicon due to the wide bandgap, making dead-time management critical. [26] [12] [27]

2.15.1 Minority Carrier Parameters

The minority carrier lifetimes in GaN are taken from published simulation data for bulk GaN power devices:

$$\tau_n = 0.7 \text{ ns}, \tau_p = 2.0 \text{ ns}$$

2.15.2 Field-Dependent Hole Mobility (Caughey-Thomas Model)

The hole mobility under the high electric field across the drift region is modelled using the Caughey-Thomas high-field saturation model.

$$\mu_p = \frac{\mu_{h0}}{\left(1 + \left(\mu_{h0} * \frac{E}{v_{h,sat}}\right)^{\beta_h}\right)^{\frac{1}{\beta_h}}} = 4.56 \frac{\text{cm}^2}{\text{V} * \text{s}}$$

Parameters: $\mu_{h0} = 70 \text{ cm}^2 / \text{V} \cdot \text{s}$ (low-field), $v_{h,sat} = 7 \times 10^6 \text{ cm/s}$, $\beta_h = 0.725$. At the average drift-region field $E = V_{DS}/W_{drift} = 1.25 \times 10^8 \text{ V/m}$. [14]

2.15.3 Diffusion Lengths and Saturation Current

Minority carrier diffusivities (Einstein Relation):

$$D_n = \mu_n V_t, D_p = \mu_p V_t$$

Minority carrier diffusion lengths:

$$L_n = \sqrt{D_n \tau_n}, L_p = \sqrt{D_p \tau_p}$$

Reverse saturation current:

$$I_0 = qA \left(\frac{D_n n_i^2}{L_n N_a} + \frac{D_p n_i^2}{L_p N_d} \right)$$

2.15.4 Forward Voltage Drop

The forward voltage is computed at the datasheet test current $I_F = 26.3 \text{ A}$:

$$V_{SD} = V_t \ln \left(\frac{I_F}{I_0} + 1 \right) = 3.271 \text{ V}$$

2.15.5 Reverse Recovery

The reverse recovery charge Q_{rr} and peak reverse current I_{rrm} are calculated from the minority carrier lifetime and the commutation speed $di/dt = 100 \text{ A}/\mu\text{s}$ (test condition):

$$Q_{rr} = I_F \tau_{eff} = 18.41 \text{ nC}$$

$$I_{rrm} = \sqrt{2Q_{rr} \frac{di}{dt}} = 1.919 \text{ A}$$

$$t_{rr} = \sqrt{\frac{2Q_{rr}}{\frac{di}{dt}}} = 19.19 \text{ ns}$$

2.16 Avalanche Analysis

Unclamped Inductive Switching (UIS) occurs when the MOSFET turns off into an inductive load without a freewheeling diode. The inductor forces V_{DS} to rise until it exceeds the device breakdown voltage V_b , whereupon avalanche conduction begins. The total energy stored in the inductor at the moment of turn-off is then dissipated inside the device. This section characterizes both the single-pulse and repetitive avalanche capability of the designed GaN MOSFET.

Unlike silicon MOSFETs, GaN devices are generally not specified or guaranteed for avalanche operation by their manufacturers. The analysis below establishes the physical limit of the device but the design should avoid entering avalanche in normal operation. [28] [29] [30]

2.16.1 Single-Pulse Avalanche Energy (E_{AS})

For pulse durations much shorter than the thermal time constant, the device can be modelled as adiabatic where all dissipated energy heats the active volume before it can flow to the package.

Maximum adiabatic avalanche energy:

$$E_{AS} = \rho_{GaN} \cdot c_p \cdot V_{active} \cdot \Delta T = 0.0578 \text{ mJ}$$

where $V_{active} = W \times L \times W_{drift}$ is the active device volume and $\Delta T = T_{j,max} - T_{case}$ is the allowable temperature rise. ρ_{GaN} is the density of GaN at 6150 kg/m³. C_p is the specific heat capacity of GaN averaged over ΔT at 470 J/kgK. [31]

2.16.2 Repetitive Avalanche Energy (E_{AR}) and Current (I_{AR})

For high-frequency switching, the device cannot cool completely between pulses. The team calculated the repetitive thermal budget based on the junction-to-case thermal resistance ($R_{th,jc}=0.5 \text{ C/W}$) to find the maximum repetitive energy and current at a test inductance of 1 μH .

$$P_{av,available} = P_{max} - P_{existing}$$

$$P_{max} = \frac{T_{j,max} - T_{case}}{R_{th,jc}}$$

$$E_{AR} = \frac{P_{av,available}}{f_{sw}} = 0.155 \frac{mJ}{cycle}$$

$$I_{AR} = \sqrt{\frac{2E_{AR}(V_b - V_{bus})}{LV_b}} = 16.84 A$$

2.17 Transconductance & MOSFET Ruggedness

Transconductance is calculated using the formula below:

$$g_m = \frac{\mu_n C_{ox} W}{L} (V_{Gs} - V_{th}) = 57.35 S \left(\frac{A}{V} \right)$$

MOSFET Ruggedness is approximated using the formula below:

$$\frac{dv}{dt} = \frac{V_{DS}}{t_f} = 75.60 \frac{V}{ns}$$

3. Performance Plots

3.1 Power Dissipation

The team plotted maximum power dissipation for varying case temperatures, T_c , using the framework outlined in 2.10.1 Maximum Continuous Drain Current. As expected, when the case temperature increases, the maximum total power dissipation decreases. Consistent with the calculations in the cited section above, the max junction temperature was set to $125^{\circ}C$ and a junction to case thermal resistance of $0.5^{\circ}C/W$ was used. See the plot in Figure 3 below.

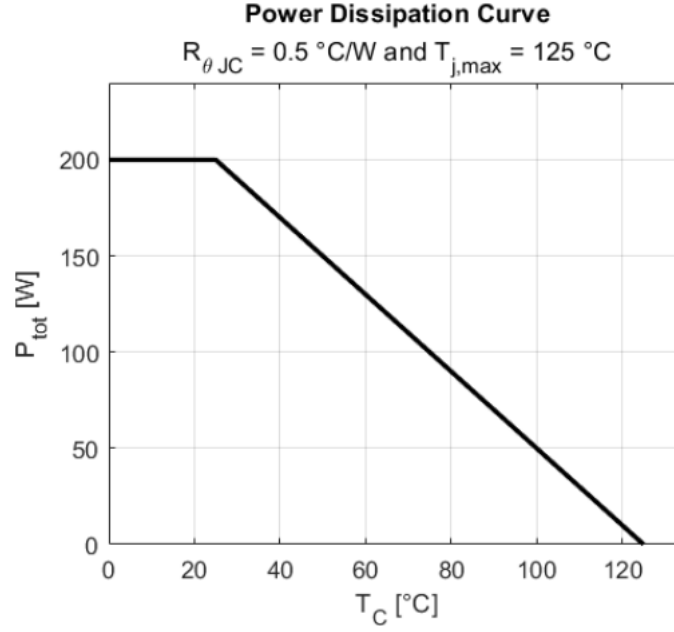


Figure 3: Power dissipation vs case temperature plot.

3.2 Transient Thermal Impedance

The team plotted the transient thermal impedance vs. pulse width to visualize how the impedance changes with both the duration of pulse width and the duty cycle. Calculations follow the formulas from Texas Instruments Application Report SLUAAT8 [19] discussed in 2.10.2 Maximum Pulsed Drain Current above. For a single, non-repetitive pulse, the one-dimensional heat diffusion model impedance formula used in the above cited section was

used: $Z_{\theta JC} = \sqrt{\frac{t_p}{t_{ref}}}$. For repetitive pulses at varying duty cycles, the thermal impedance was modeled as a linear interpolation between the single-pulse transient response and the

steady-state limit of 1: $Z_{\theta JC, repetitive} = \min\left(1, D + (1 - D) * \sqrt{\frac{t_p}{t_{ref}}}\right)$. This approximation would be most accurate for low to moderate duty cycles but serves as a conservative estimate for this design. Additionally, it does not account for thermal accumulation effects over multiple switching cycles. See Figure 4 below.

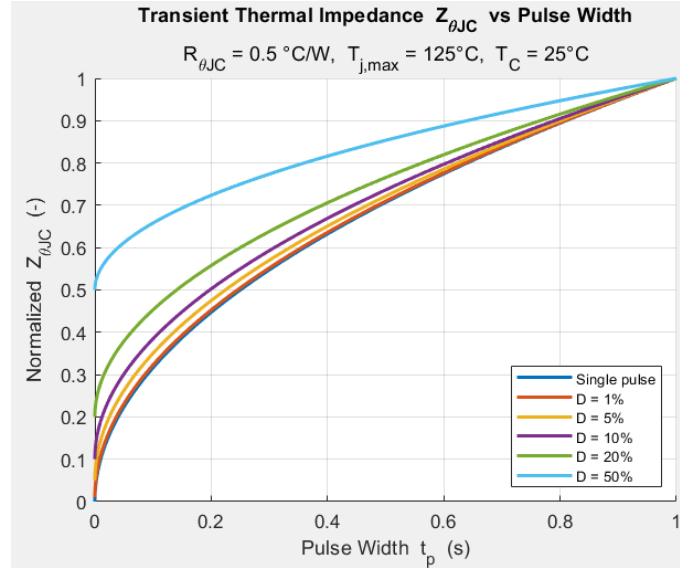
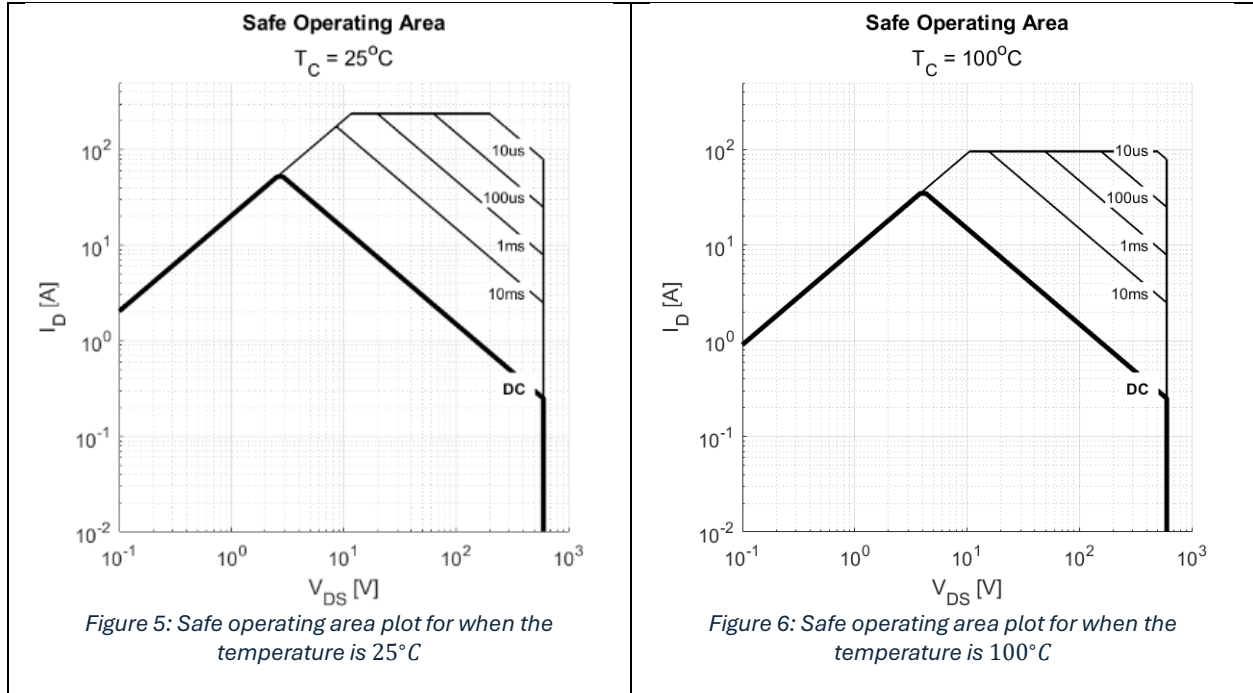


Figure 4: Transient Thermal Impedance vs Pulse Width

3.3 Safe Operating Area

The team plotted the safe operating area at 25°C and 100°C to visualize the boundary conditions the designed device can operate in without failing. These graphs were made in MATLAB combining calculations the team had made for boundary conditions. Each line has either a DC label or a pulse duration label. These lines correspond to the boundary conditions if the MOSFET was operated at a constant voltage (DC) or single pulse. The horizontal boundary line is the absolute maximum current the MOSFET can handle before failure at a specific pulse, the left slope is $R_{DS(on)}$, and the right descending slope is the maximum power/thermal limit. See both plots in Figure 5 and Figure 6.



3.4 I_D vs V_{DS}

A MATLAB script was used to develop I_D vs. V_{DS} plots at varying V_{GS} values at a junction temperature of 25 °C and 100°C, as shown in Figure 7 and Figure 8 respectively. V_{GS} values were varied to range from 1 V above the threshold voltage (2.7 V) to the maximum gate drive voltage (15 V) in steps. To create this plot, for each V_{GS} value, the script would sweep through the range of V_{DS} values, and at each point, identify the state of the MOSFET and apply the appropriate current formula. In the cut-off region, where $V_{GS} \leq V_{th}$, $I_D = 0$. In the linear region, where $V_{DS} < V_{GS} - V_{th}$, $I_D = \frac{w}{L} \mu_n C_{ox} * \left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right)$. Lastly, in the saturation region, where $V_{DS} \geq V_{GS} - V_{th}$, $I_D = \frac{1}{2} \frac{w}{L} \mu_n C_{ox} (V_{GS} - V_{th})^2$. As seen in the figures below, these plots help represent the behaviour of the theoretical drain currents based on the device design, however, the maximum operational continuous drain current is limited to the thermally limited value derived and discussed above.

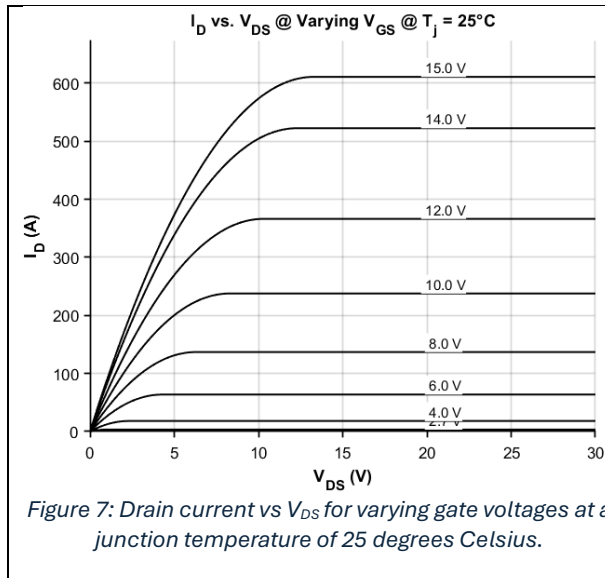


Figure 7: Drain current vs V_{DS} for varying gate voltages at a junction temperature of 25 degrees Celsius.

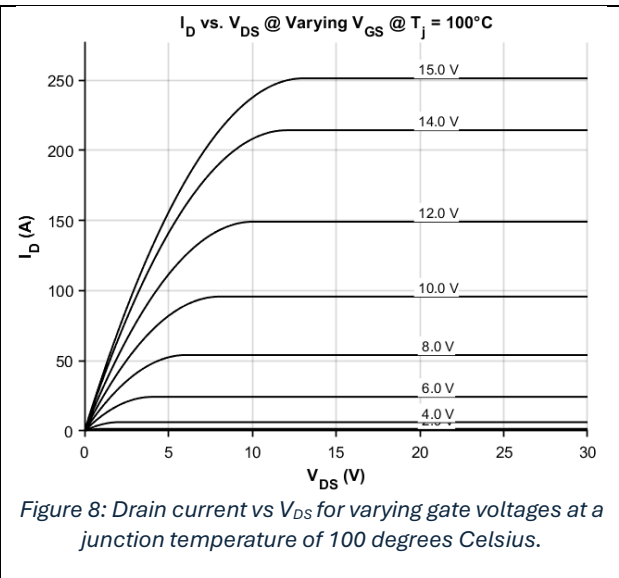


Figure 8: Drain current vs V_{DS} for varying gate voltages at a junction temperature of 100 degrees Celsius.

3.5 $R_{DS(on)}$ vs Drain Current

The team plotted the $R_{DS(on)}$ resistance vs. increasing drain currents at 100 degrees Celsius to visualize how the on resistance increases with current. Six different lines were made increasing V_{GS} to show the impact of this parameter increasing. See Figure 9 below.

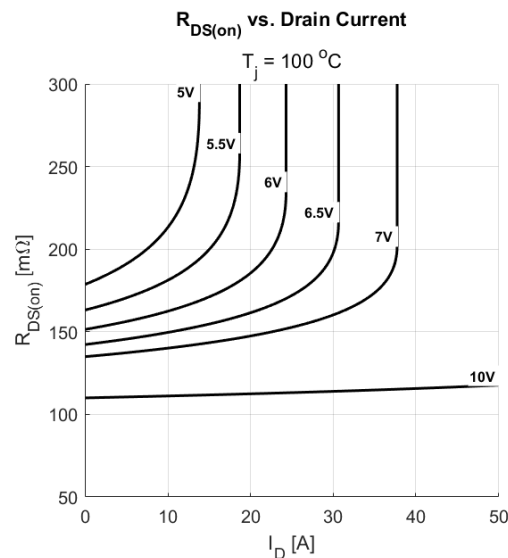


Figure 9: $R_{DS(on)}$ vs Drain Current Plot

3.6 $R_{DS(on)}$ vs Temperature

The team plotted the on resistance versus the junction temperature T_j to visual how the contact resistance increases with temperature. The resistance calculations were made following the equations discussed in the 2.9 On-State Resistance section.

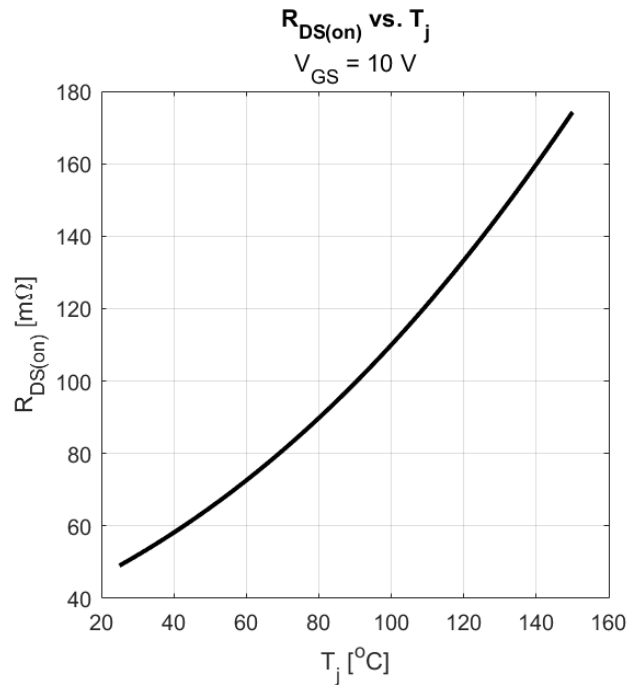


Figure 10: $R_{ds(on)}$ vs Junction Temperature Plot

3.7 Recovery Current and Time vs Commutation Speed

Figure 11 illustrates the transistor's efficiency during fast switching cycles. It plots the peak recovery current and the time it takes to clear that charge against how fast the device is switched.

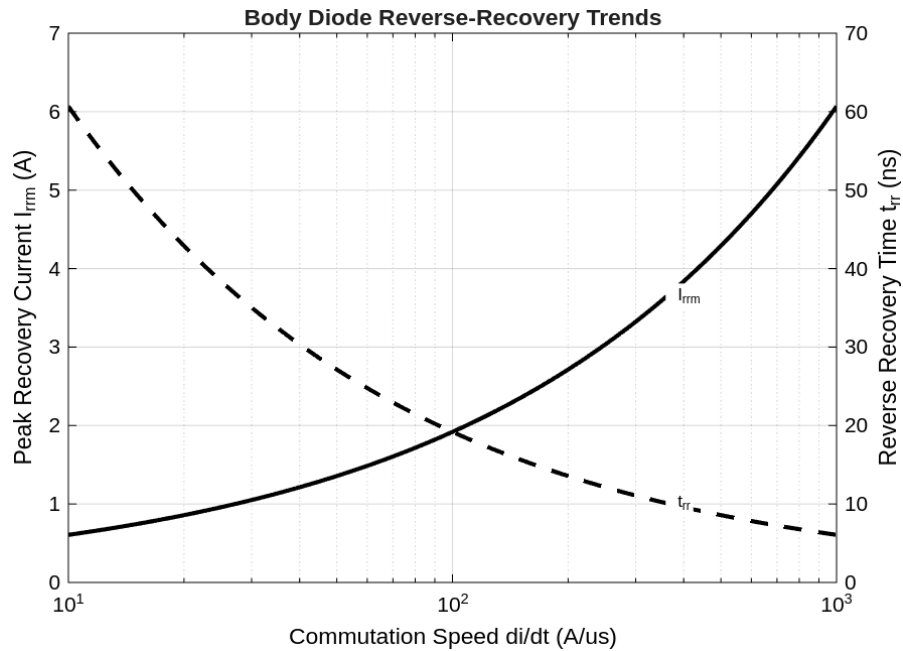


Figure 11: Recovery current and time vs Commutation speed plot.

3.9 Avalanche Energy vs Junction Temperature

Here is mapped the device's ability to survive high-voltage spikes as it heats up. The energy naturally drops at higher temperatures.

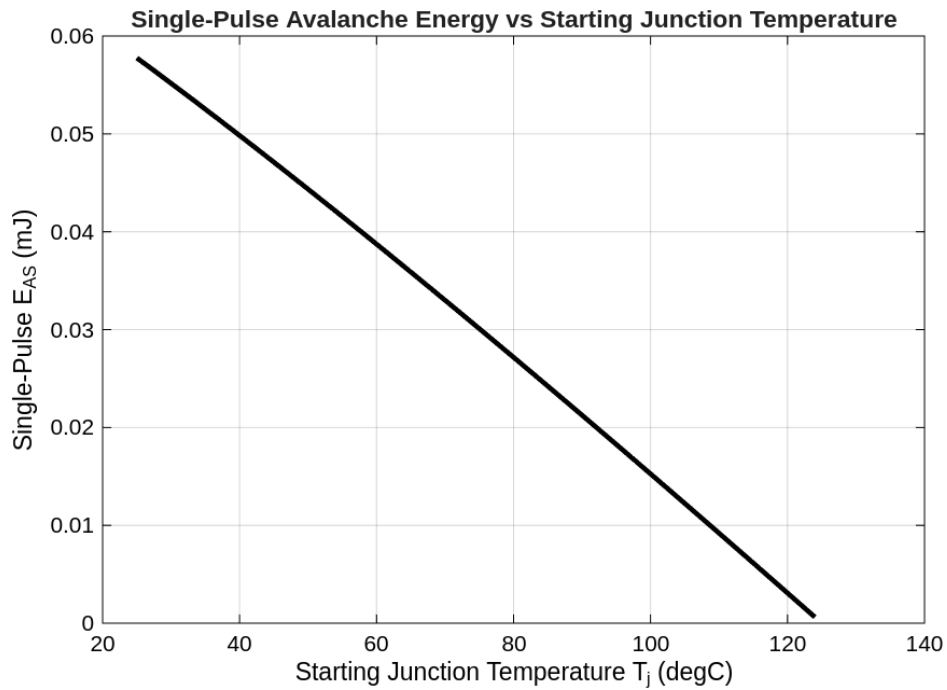


Figure 12: E_{AS} vs Junction temperature plot.

3.8 Transfer characteristics (I_D vs V_{GS})

Figure 13 shows how much drain current flows through the device for a given gate voltage. It highlights the threshold voltage where the transistor turns on and demonstrates the steep slope of the curve as the Miller plateau is approached. That steep slope means the device can rapidly switch massive currents on and off with minimal gate drive effort, which is essential for high-speed operation.

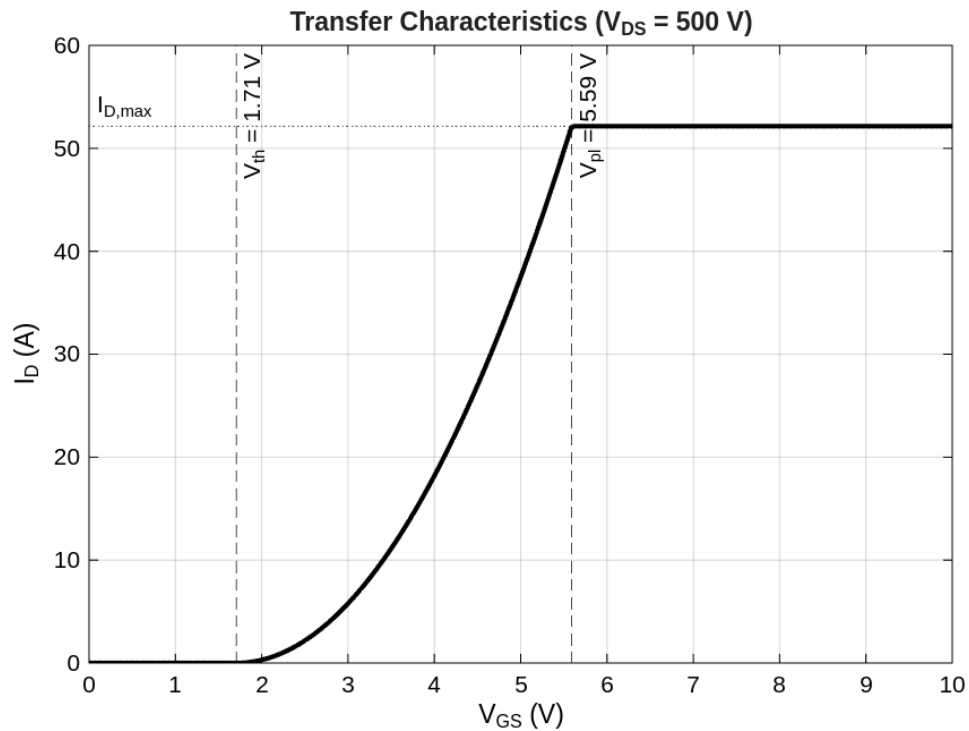


Figure 13: I_D vs V_{GS} plot.

4. Conclusion

This report presented the team's design of a high-speed high-power GaN LDMOS transistor rated for a 600 V breakdown voltage and a 52.14 A continuous drain current. The design proceeded from material selection through to full electrical, thermal, and avalanche characterization. In addition to the bulk LDMOS architecture, the design explores a multi-finger gate layout, which involves fifty 1mm fingers wired in parallel through interconnect metallization, as the means of addressing the gate-resistance and current-density problems that are more commonly handled in GaN by adopting a HEMT topology.

Gallium nitride was selected as the substrate over silicon and 4H-SiC. Its wide bandgap and critical field near 3.3 MV/cm permit a drift region an order of magnitude shorter than silicon would require at the same blocking voltage, while its electron mobility exceeds that of 4H-

SiC by roughly a factor of two, which is necessary to meet the high-speed requirement. A lateral DMOS topology was adopted to decouple the threshold voltage from the breakdown voltage, a decoupling that a conventional planar NMOS cannot provide at this voltage.

The most significant departure from conventional practice is the choice of acceptor. P-type doping in GaN is the long-standing weak point of the material system. Magnesium, the industry-standard acceptor, suffers from an ionization energy of roughly 170 meV, which places it well above the thermal energy at room temperature and leaves only a small fraction of the nominal doping electrically active. At the raw concentration used here, magnesium would be insufficiently ionized at 300 K, and most atoms would contribute nothing to the channel.

The team adopted beryllium in place of magnesium. Beryllium has a lower ionization energy than magnesium, which raises the room-temperature ionization fraction substantially. This is the primary lever by which the design achieves a usable p-body at moderate raw doping, and the effective channel doping used throughout the threshold-voltage calculation is a direct consequence of this choice.

Beryllium doping has its own issues. Beryllium is toxic to work with, and free Be acceptors in GaN are known to be compensated by unintentional donors introduced during growth, which would offset much of the ionization advantage if left unaddressed. The team's design circumvents this through the hydrogen-passivation route described by Demchenko and Reshchikov, in which beryllium is introduced into the lattice bound to hydrogen as Be-H complexes during growth. In this form the Be is electrically inactive and does not pin the Fermi level in a way that promotes compensating-donor incorporation, so the growth proceeds cleanly. The hydrogen is then liberated by a post-growth thermal anneal, leaving behind active Be acceptors in the intended concentration. The result is a p-type GaN body in which the doping is set deliberately by the activation step rather than fought against by compensation during growth. This is a novel research-stage technique rather than a production technique.

With this acceptor choice established, the remainder of the design follows. The drift region is dimensioned with a safety margin above the critical-field minimum. The threshold voltage places the device firmly in enhancement mode, and the on-state resistance is dominated by the drift region with a smaller contribution from the channel and a negligible contribution from the ohmic contacts. Under the thermal limit set by the TO-247 package, the continuous current and the peak pulsed current at 100 μ s maintain roughly the four-to-one ratio expected of commercial power MOSFETs in this class. The Miller plateau lies between threshold and the gate drive voltage, confirming a well-defined switching region. Switching analysis yields nanosecond-scale rise and fall times and a theoretical switching frequency

in the tens of megahertz, but the thermally-limited operating frequency falls in the hundreds of kilohertz, consistent with commercial GaN power MOSFETs in the same voltage class. The body diode exhibits the elevated forward voltage characteristic of wide-bandgap devices, with fast reverse recovery. Avalanche analysis is included for completeness despite GaN devices not being rated for avalanche by manufacturers.

In conclusion, these results describe a device that meets its specification in every dimension examined. The architecture, material choice, dopant chemistry, and geometry are mutually consistent. The wide bandgap of GaN permits a short drift region; the short drift region permits a low on-resistance at the rated blocking voltage; the low on-resistance permits the target continuous current within the package thermal envelope; the Be-H doping route permits a usable p-body at doping levels that Mg could not deliver without compensation; and the multi-finger layout permits the total gate width demanded by the current rating without the gate-RC penalty that would otherwise force the design toward a HEMT. Modifications in any one of these choices would propagate through the others, and the design closes properly.

References

- [1] Offe, "Basic Parameters for Wurtzite Crystal Structure," [Online]. Available: <https://www.ioffe.ru/SVA/NSM/Semicond/GaN/basic.html>. [Accessed 2 April 2026].
- [2] EEsemi, "Properties of Si, Ge, and GaAs at 300K," EEsemi, 2004. [Online]. Available: <https://eesemi.com/sigegaas.htm>. [Accessed 3 April 2026].
- [3] M. A. R. Demchenko, "Passivation of beryllium acceptor in GaN and a possible route for p-type doping," Department of Physics, Virginia Commonwealth University.
- [4] Infineon, "CoolMOS C6 650V CoolMOS™ C6 Power Transistor," Infineon, 15 March 2011. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/49/infineon-ipw65r070c6-ds-en.pdf>.
- [5] Infineon, "IGT65R025D2 Final Datasheet," Infineon, 5 March 2026. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/49/infineon-igt65r025d2-datasheet-en.pdf?fileId=8ac78c8c8eeb092c018f0a57bd2f0307>.
- [6] A. Logotheti, "Low Resistivity n-type GaN Ohmic Contacts on GaN Substrates," Wiley, 18 November 2024. [Online]. Available: <https://onlinelibrary.wiley.com/doi/full/10.1002/pssa.202400692>.
- [7] A. M. Bergemont, "Multi-Finger MOS Transistor With Reduced Gate Resistance". United States Patent 5,789,791, 4 August 1998.
- [8] NDT Supply, "Conductivity and Resistivity Values for Aluminum & Alloys," NDT Supply, March 2002. [Online]. Available: https://content.ndtsupply.com/media/Conductivity_Al%20Reference%20Chart.pdf.
- [9] Synopsys, "Gate Resistance in IC Design Flow," Synopsys, 22 January 2026. [Online]. Available: <https://www.synopsys.com/blogs/chip-design/gate-resistance-ic-design-flow.html>.
- [10] S. M. Sze and S. S. Ng, Physics of Semiconductor Devices, 3rd ed. Wiley, 2007.

- [11] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*, Springer, 2008.
- [12] G. Sabui, P. J. Parbrook and M. Arredondo, "Modeling and simulation of bulk gallium nitride power semiconductor devices," *AIP Advances*, May 2016. [Online]. Available: https://www.researchgate.net/publication/301828661_Modeling_and_simulation_of_bulk_gallium_nitride_power_semiconductor_devices. [Accessed 3 April 2026].
- [13] T. T. Mnatsakanov, "Carrier mobility model for GaN," *Solid-State Electronics*, vol. 47, pp. 111-115, 2003.
- [14] P. Fiorenza, G. Greco, F. Iucolano, A. Patti and F. Roccaforte, "Channel mobility in GaN hybrid MOS-HEMT using SiO₂ as gate insulator," *IEEE Transaction on Electron Devices*, 2017.
- [15] M. Meneghini, C. De Santi, I. Abid, M. Buffolo, M. Cioni, R. Abdul Khadar, L. Nela, N. Zagni, A. Chini, F. Medjdoub, G. Meneghesso, G. Verzellesi, E. Zanoni and E. Matioli, "GaN-based power devices: Physics, reliability, and perspectives," 8 November 2021. [Online]. Available: <https://pubs.aip.org/aip/jap/article/130/18/181101/1025623/GaN-based-power-devices-Physics-reliability-and>.
- [16] T. Hashizume, "Interface properties of GaN MOS structures," *Appl. Phys. Lett.*, pp. 2952-2954, 2003.
- [17] S. Zhonghao, H. Huang, N. Sun, P. Tao, C. Zhao and Y. C. Liang, "A Novel GaN Metal-Insulator-Semiconductor High Electron Mobility Transistor Featuring Vertical Gate Structure," 5 December 2019. [Online]. Available: <https://www.mdpi.com/2072-666X/10/12/848>.
- [18] V. N. Brudnyi, "BN, AlN, GaN, InN: Charge Neutrality Level, Surface, Interfaces, Doping," April 2017. [Online]. Available: https://www.researchgate.net/publication/315887149_BN_AlN_GaN_InN_Charge_Neutrality_Level_Surface_Interfaces_Doping.
- [19] J. Wallace, "Using MOSFET Transient Thermal Impedance Curves in Your Design," December 2023. [Online]. Available: [https://www.ti.com/lit/an/sluaat8/sluaat8.pdf?ts=1775631107782#:~:text=The%20operating%20conditions%20including%20gate,\(2\)](https://www.ti.com/lit/an/sluaat8/sluaat8.pdf?ts=1775631107782#:~:text=The%20operating%20conditions%20including%20gate,(2)).

- [20] A. Harb and I. Batarseh, "Gate charge and Miller plateau analysis," in *Power Electronics: Circuit Analysis and Design*, Springer, 2018.
- [21] Texas Instruments, "Understanding Power MOSFET Switching Performance," 2013. [Online].
- [22] Y. Tsividis and C. McAndrew, *Operation and Modeling of the MOS Transistor*, Oxford Univ. Press, 2011.
- [23] Vishay, "Power MOSFET Basics: Understanding Gate Charge and Using it to Access Switching Performance," 16 February 2016. [Online]. Available: <https://www.vishay.com/docs/73217/an608a.pdf>.
- [24] Q. Kabashi, N. Caka and M. Zabeli, "Impact of MOSFET parameters on its parasitic capacitances," 2007. [Online]. Available: https://www.academia.edu/3030302/Impact_of_MOSFET_parameters_on_its_parasitic_capacitances.
- [25] Toshiba Electronic Devices & Storage Corporation, "MOSFET Gate Drive Circuit Application Note," 26 July 2018. [Online]. Available: https://toshiba.semicon-storage.com/info/application_note_en_20180726_AKX00068.pdf?did=59460.
- [26] O. Ambacher, Two-dimensional electron gases induced by spontaneous and piezoelectric polarization, *J. Appl. Phys.*, 1999.
- [27] D. Caughey and R. Thomas, "Carrier mobilities in silicon empirically related to doping and field," in *IEEE*, 1967.
- [28] Vishay Siliconix, "Power MOSFET in UIS/Avalanche Applications," 22 December 2008. [Online]. Available: <https://www.vishay.com/docs/64717/an912.pdf>.
- [29] Nexperia Assets, "Power MOSFET single-shot and repetitive avalanche ruggedness rating," 12 February 2026. [Online]. Available: <https://assets.nexperia.com/documents/application-note/AN10273.pdf>.
- [30] A. Kumar, S. Parashar, J. Baliga and S. Bhattacharya, "Single Shot Avalanche Energy Characterization of 10kV, 10A 4H-SiC MOSFETs," 2018. [Online]. Available: <https://www.freedm.ncsu.edu/wp-content/uploads/2018/03/A.-Kumar-APEC-2018.pdf>.

- [31] K. Xu, H. Gu, Y. Pei, X. Yang and A. A. Al-Saman, "An analytical physics-based model for low-field mobility in nitride high-electron-mobility transistor considering the momentum relaxation and two-dimensional electronic gas effective mass," 9 October 2025. [Online]. Available: <https://pubs.aip.org/aip/jap/article/138/14/144502/3367200/An-analytical-physics-based-model-for-low-field>.
- [32] T. Mnatsakanov and M. E. Levinshtein, "Carrier mobility model for GaN," ResearchGate, January 2003. [Online]. Available: https://www.researchgate.net/publication/222028285_Carrier_mobility_model_for_GaN. [Accessed 3 April 2026].
- [33] "Basic Parameters of Gallium Nitride (BN)," [Online]. Available: <https://www.ioffe.ru/SVA/NSM/Semicond/GaN/basic.html>. [Accessed 13 April 2026].

Appendix A: MATLAB Calculations Output

```
-----  
Model Results  
-----
```

```
Channel mobility          mu_n = 1201.5 cm2/Vs  
Oxide capacitor density  Cox = 1.1510e-03 F/m2  
Oxide Electric Field     E_ox = 3.3333 MV/cm  
Max depletion width      x_dT = 0.2520 um  
Threshold voltage        Vth = 1.7066 V  
Plateau voltage          Vgs_pl = 5.5902 V  
Drift width              W_drift = 4.0000 um  
Drain current continuous Id = 52.1449 A  
On-resistance            R_on = 0.0490 ohm  
Transit frequency        fT = 158.5929 GHz  
Rise time                tr = 8.0630 ns  
Fall time                tf = 6.6140 ns  
On Time                  tdon = 12.6440 ns  
Off Time                  toff = 12.7893 ns  
Switching freq. Max      f_sw = 24.9313 MHz  
Switching freq. Op       f_sw = 0.6912 MHz  
Transconductance         gm = 57.3478 S (A/V)  
>> |
```

1. Drift Region Calculations:

Background doping $N_b = 3.6892 \times 10^{22} \text{ m}^{-3}$ ($3.6892 \times 10^{16} \text{ cm}^{-3}$)

Drift region width $W_{\text{drift}} = 4.0000 \times 10^{-6} \text{ m}$ ($4.0000 \text{ } \mu\text{m}$)

2. Band Gap, Intrinsic Carrier Concentration, and Mobility Calculations ($T = 2.9815 \times 10^2 \text{ K}$)

Thermal voltage $V_t = 0.02570 \text{ V}$

GaN bandgap $E_g(T) = 3.4354 \text{ eV}$

Intrinsic carrier concentration $n_i(T) = 1.2647 \times 10^{-4} \text{ m}^{-3}$ ($1.2647 \times 10^{-10} \text{ cm}^{-3}$)

Channel mobility $\mu_n(\text{ch}) = 1201.52 \text{ cm}^2/\text{Vs}$

Drift region mobility $\mu_n(\text{dr}) = 1347.68 \text{ cm}^2/\text{Vs}$

3. MOS Capacitor and Threshold Voltage

Oxide capacitance $C_{\text{ox}} = 1.1510 \times 10^{-3} \text{ F/m}^2$

Oxide Electric Field $E_{\text{ox}} = 3.3333 \text{ MV/cm}$

Fermi potential $\phi_{\text{fp}} = 1.5734 \text{ V}$

Max depletion width $x_{\text{dT}} = 2.5196 \times 10^{-7} \text{ m}$ ($0.2520 \text{ } \mu\text{m}$)

Max depletion charge $Q_{\text{SD}} = 1.9684 \times 10^{-3} \text{ C/m}^2$

Work function difference $\phi_{\text{ms}} = -3.0111 \text{ V}$

Threshold voltage $V_{\text{th}} = 1.7066 \text{ V}$

[NOTE] Device is ENHANCEMENT-MODE ($V_{\text{th}} > 0$)

4: MOSFET I-V Characteristics Operating region: Saturation

Drain current $I_d = 237.8035 \text{ A}$

Transit frequency $f_T = 1.5859 \times 10^{11} \text{ Hz}$ (158.5929 GHz)

5: On Resistance

Contact Resistance $R_c = 0.7331 \text{ mohm}$

Channel Resistance $R_{\text{ch}} = 17.4375 \text{ mohm}$

Drift Resistance $R_{\text{drift}} = 30.1324 \text{ mohm}$

Total On Resistance $R_{\text{on}} = 49.0360 \text{ mohm}$

Transconductance $g_m = 57.3478 \text{ S}$ (A/V)

6: Thermal Current Limits Calculations (TI SLUAAT8)

$R_{ds(on)}$ @ T_{j_max} = 73.5540 mohm

Max power diss = 200.00 W

Continuous I_{D_max} = 52.14 A

Peak I_{DM} (100us) = 521.45 A

7: Gate Plateau Voltage Calculations

Plateau voltage V_{gs_pl} = 5.5902 V

8: Gate Capacitances Calculations

Saturation: C_{gs} = 4.9878e-11 F, C_{gd} = 1.1510e-11 F

Triode: C_{gs} = 4.0286e-11 F, C_{gd} = 4.0286e-11 F

Cutoff: C_{gs} = 1.1510e-11 F, C_{gd} = 1.1510e-11 F

9: Switching times calculations

Fall time t_f = 6.6140e-09 s (6.6140 ns)

Rise time t_r = 8.0630e-09 s (8.0630 ns)

On Delay t_{don} = 1.2644e-08 s (12.6440 ns)

Off Delay t_{off} = 1.2789e-08 s (12.7893 ns)

Switching frequency f_{sw} = 2.4931e+07 Hz (24.9313 MHz)

10: Gate Charge Parameters

Charge to V_{th} Q_{gs1} = 1.0477e-10 C (0.1048 nC)

Charge V_{th} to P_{lt} . Q_{gs2} = 2.3840e-10 C (0.2384 nC)

Total Gate-Source Q_{gs} = 3.4317e-10 C (0.3432 nC)

Miller Charge Q_{gd} = 5.7551e-09 C (5.7551 nC)

Total Gate Charge Q_g = 6.0983e-09 C (6.0983 nC)

Input Capacitance C_{iss} = 6.1388e-11 F (0.0614 nF)

```

11: p-n Junction Calculations
Built-in potential phi_0 = 3.2836 V
Junction area Aj = 1.2598e-08 m2
Zero-bias junction cap Cj0 = 3.0543e-04 F/m2
Keq (0 -> -500V) Keq = 0.149473

Cds = 5.7513e-13 F
C_oss = 1.2085e-11 F

12: dv/dt Calculation
Estimated dv/dt = 7.5597e+10 V/s (75.60 V/ns)

13: Power Analysis Calculations
Conduction Loss P_cond = 66.6667 W
Thermal Limit Frequency (fsw_p): 691.19 kHz
Switching Loss P_sw = 132.2470 W
Energy per cycle E_coss = 1.5107e-06 J
Coss Loss P_coss = 1.0442 W
Gate Drive Loss P_gate = 0.0042 W
TOTAL POWER LOSS P_tot = 199.9621 W

```

Appendix B: MATLAB Calculations

```

clear; clc; close all;

% Physical constants
q = 1.602e-19; % Elementary charge [C]
k = 1.381e-23; % Boltzmann constant [J/K]
eps0 = 8.854e-12; % Permittivity of free space [F/m]
h = 6.626e-34; % Planck's constant [Js]
m_0 = 9.1093837 * 10^(-31); % electron mass [kg]

% Independent parameters
T = 298.15; % Operating temperature [K] - Typ. 298.15 (25), Max 398.15 (125), Min 243.15(-30)

R_g = 6; % Gate resistance [ohm] - Intrinsic is 135 mohm, but 6 is used to account for gate drive circuit resistance
V_ds = 500; % Drain-source voltage [V]
V_g = 10; % Gate drive voltage [V] operating at 10, max 15 (due to Eox)
Vb = 600; % Breakdown voltage [V]

```

```

% Geometry
W = 50e-3; % Gate width [m] 50 mm
L = 1e-6; % Gate length [m] 1 um
t_ox = 30e-9; % Gate oxide thickness [m] (30 nm)
L_ov = 0.2e-6; % Gate overlap length [m] (0.2 um)
c_w = W; % contact width
c_l = 4e-6; % contact length
H = 5e-6; % height of device

% Doping Concentrations
Nd = 1e19 * 1e6; % Donor doping
Na_raw = 8e16 * 1e6; % Acceptor doping

% GaN Constants
Nv_300 = 4.6e19 * 1e6; % Effective density of states in valence band
Nc_300 = 2.3 * 10^24; % Effective density of states in the conduction band
E_crit = 3e6 * 1e2; % Critical electric field [V/m]
er_GaN = 8.9; % Relative permittivity of GaN
e_s = er_GaN * eps0; % Absolute permittivity of GaN [F/m]
chi_GaN = 4.1; % Electron affinity of GaN [eV]
Eg0_GaN = 3.507; % Bandgap at 0 K [eV]
alpha_V = 0.909e-3; % Varshni alpha for GaN [eV/K]
beta_V = 830; % Varshni beta for GaN [K]
ni_300 = 1.9e-10 * 1e6; % Intrinsic carrier concentration at 300 K [m-3]
m_n = 0.2 * m_0; % effective electron mass in GaN [kg]

% SiO2 Constants
er_ox = 3.9; % Relative permittivity of SiO2
e_ox = er_ox * eps0; % Absolute permittivity of SiO2 [F/m]

% Contact constants
Qss_p = 1e11 * 1e4; % Interface charge density [m-2]
phi_m = 4.38; % Gate metal (Al) work function [eV]
phi_Ti = 4.33; % Titanium work function

% Beryllium ionization energy in GaN
Be_ion = 116e-3;

% Mobility Model Parameters
NREF = 3e17 * 1e6; % Reference doping [m-3]
mu_max = 1600 * 1e-4; % Max electron mobility [m2/V.s]
mu_min = 100 * 1e-4; % Min electron mobility [m2/V.s]
beta1_m = 3; % Temperature exponent for mumax
gamma_m = 4.4; % Temperature exponent for NREF
lambda1 = 0.7; % Doping exponent

% Thermal current parameters
Tj_max = 125; % Max junction temperature [degrees C]
T_case = 25; % Case temperature [degrees C]
Rth_jc = 0.5; % Junction-to-case thermal resistance [degrees C/W]
alpha = 0.005; % Rds(on) temp coefficient [1/degrees C]

% Tweaking Device Parameters for min and max variation
% t_ox = 1.1*t_ox;
% Na = 0.9*Na;

```

```

% % Qss_p = 1.2*Qss_p;
% Nd = 0.9*Nd;
% L = 1.05*L;

% Temperature effects on constants + ionization calculation
Nv = Nv_300 * (T/300)^(3/2);
Nc = Nc_300 * (T/300)^(3/2);

Fraction_unionized = 1 / (1 + (Nv/(4*Na_raw) * exp((-Be_ion)/(k/q*T))));
Na = Na_raw * (1 - Fraction_unionized);

% Drift Region Calculations
fprintf('1. Drift Region Calculations:\n');

% Background doping from breakdown condition
Nb = e_s * E_crit^2 / (2 * q * Vb); % [m-3]
fprintf('Background doping Nb = %.4e m-3 (%.4e cm-3)\n', Nb, Nb/1e6);

% Drift region width
% W_drift = 2*Vb / Ecrit
W_drift = 2 * Vb / E_crit; % [m]
fprintf('Drift region width W_drift= %.4e m (%.4f μm)\n\n', W_drift, W_drift*1e6);

% Effect of temperature on Eg, ni, and mobilities
fprintf('2. Band Gap, Intrinsic Carrier Concentration, and Mobility Calculations\n\n');
(T=0.4e K) \n', T);

% Thermal voltage
V_t = k * T / q; % [V]
fprintf('Thermal voltage V_t = %.5f V\n', V_t);

% GaN bandgap via Varshni equation: Eg(T) = Eg(0) - alpha *T^2/(beta+T)
Eg_T = Eg0_GaN - (alpha_V * T^2) / (beta_V + T); % [eV]
Eg_300 = Eg0_GaN - (alpha_V * 300^2) / (beta_V + 300);
fprintf('GaN bandgap Eg(T) = %.4f eV\n', Eg_T);

% Intrinsic carrier concentration
ni = ni_300 * (T/300)^(3/2) * exp(Eg_300 / (2*0.02585)) * exp(-Eg_T / (2*V_t));
fprintf('Intrinsic carrier concentration ni(T) = %.4e m-3 %.4e cm-3\n', ni, ni/1e6);

% Electron mobility in channel
mu_n_ch = mu_min + (mu_max*(300/T)^beta1_m - mu_min) / ...
(1 + (Na_raw/NREF * (T/300)^gamma_m)^lambda1);
fprintf('Channel mobility mu_n(ch) = %.2f cm2/Vs\n', mu_n_ch*1e4);

% Electron mobility in drift region
mu_n_dr = mu_min + (mu_max*(300/T)^beta1_m - mu_min) / ...
(1 + (Nb/NREF * (T/300)^gamma_m)^lambda1);
fprintf('Drift region mobility mu_n(dr) = %.2f cm2/Vs\n\n', mu_n_dr*1e4);

% Use channel mobility
mu_n = mu_n_ch;

% MOS capacitor and threshold voltage calculations

```

```

fprintf('3. MOS Capacitor and Threshold Voltage\n');

% Oxide capacitance per unit area
C_ox = e_ox / t_ox; % [F/m2]
E_ox = (V_g / (t_ox*1e9))* 10;
fprintf('Oxide capacitance C_ox = %.4e F/m2\n', C_ox);
fprintf('Oxide Electric Field E_ox = %.4f MV/cm\n', E_ox);

% Fermi potential
phi_fp = V_t * log(Na / ni); % [V]
fprintf('Fermi potential phi_fp = %.4f V\n', phi_fp);

% Maximum depletion width
x_dT = sqrt((4 * e_s * phi_fp) / (q * Na)); % [m]
fprintf('Max depletion width x_dT = %.4e m (%.4f mum)\n', x_dT, x_dT*1e6);

% Maximum depletion charge
Q_SD_max = q * Na * x_dT; % [C/m2]
fprintf('Max depletion charge QSD = %.4e C/m2\n', Q_SD_max);

% Metal-semiconductor work function difference
phi_ms = phi_m - (chi_GaN + Eg_T/2 + phi_fp); % [V]
fprintf('Work function difference phi_ms = %.4f V\n', phi_ms);

% Threshold voltage
V_th = Q_SD_max/C_ox - (q*Qss_p)/C_ox + phi_ms + 2*phi_fp;
fprintf('Threshold voltage Vth = %.4f V\n\n', V_th);

if V_th < 0
    fprintf('[NOTE] Device is DEPLETION-MODE (Vth < 0)\n\n');
else
    fprintf('[NOTE] Device is ENHANCEMENT-MODE (Vth > 0)\n\n');
end

% MOSFET I-V Characteristics
fprintf('4: MOSFET I-V Characteristics');

V_gs = V_g;
V_ds_sat = V_gs - V_th; % Saturation onset voltage
lambda = 0; % Channel-length modulation coefficient

% Determine operating region
if V_gs <= V_th
    region = 'Cutoff';
    I_d = 0;
elseif V_ds < V_ds_sat
    region = 'Triode';
    I_d = W * mu_n * C_ox * (2*(V_gs-V_th)*V_ds - V_ds^2) / (2*L);
else
    region = 'Saturation';
    I_d = W * mu_n * C_ox * (V_gs - V_th)^2 * (1 + lambda*V_ds) / (2*L);
end

fprintf('Operating region: %s\n', region);
fprintf('Drain current Id = %.4f A\n', I_d);

```

```

% Transit frequency
f_T = mu_n * (V_gs - V_th) / (2 * pi * L^2);
fprintf('Transit frequency f_T = %.4e Hz (%.4f GHz)\n\n', f_T, f_T*1e-9);

% On resistance calculation
fprintf('5: On Resistance\n');

phi_B = phi_Ti - chi_GaN;
E_00 = (q/2) * (h / (2*pi)) * sqrt(Nd/(e_s * m_n));
A_star = (4 * pi * q * m_n * k^2) / h^3;
phi_n = (k*T / q) * log(Nc/Nd);
rho_c_1 = k^2 / (q * A_star);
rho_c_2 = 1 / sqrt(pi * q * (phi_B + phi_n) * E_00);
rho_c_3 = cosh(E_00 / (k * T));
rho_c_4 = sqrt(coth(E_00 / (k * T)));
rho_c_5 = exp(((phi_B + phi_n) * q / (E_00 * coth(E_00 / (k * T)))) - (phi_n * q / (k * T)));
rho_c = rho_c_1 * rho_c_2 * rho_c_3 * rho_c_4 * rho_c_5;
R_c = rho_c / (c_l * c_w);

% channel resistance
R_ch = (L/W) * (1/(mu_n_ch * C_ox * (V_g-V_th)));
W_drift = (2*Vb)/E_crit;
R_drift = (W_drift*1.5) / (q * mu_n_dr * Nb * W * H);
R_on = 2 * R_c + R_ch + R_drift;
fprintf('Contact Resistance R_c = %.4f mohm\n', R_c * 1e3);
fprintf('Channel Resistance R_ch = %.4f mohm\n', R_ch * 1e3);
fprintf('Drift Resistance R_drift = %.4f mohm\n', R_drift * 1e3);
fprintf('Total On Resistance R_on = %.4f mohm\n\n', R_on * 1e3);

gm = mu_n * C_ox * (W/L) * (V_g - V_th);
fprintf('Transconductance gm = %.4f S (A/V)\n\n', gm);

% Thermal current limits following TI SLUAAT8
fprintf('6: Thermal Current Limits Calculations (TI SLUAAT8)\n');

norm_factor = 1 + alpha * (Tj_max - 25);
Rds_on_Tjmax = R_on * norm_factor;

P_max = (Tj_max - T_case) / Rth_jc; % Max power dissipation
I_D_max = sqrt(P_max / Rds_on_Tjmax); % Max continuous drain current

t_ref = 1.0;
Zth_single_pulse = @(tp) min(sqrt(tp / t_ref), 1.0);
Zth_repetitive = @(tp, D) min(D + (1-D)*sqrt(tp / t_ref), 1.0);

% Peak drain current
% I_DM: 100 mus single pulse
tp_IDM = 100e-6;
Zth_IDM = Zth_single_pulse(tp_IDM);
I_DM = sqrt((Tj_max - T_case) / (Zth_IDM * Rth_jc * Rds_on_Tjmax));

fprintf('Rds(on) @ Tj_max = %.4f mohm\n', Rds_on_Tjmax*1e3);
fprintf('Max power diss = %.2f W\n', P_max);

```

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fprintf('Continuous ID_max = %.2f A\n', I_D_max);
fprintf('Peak I_DM (100us) = %.2f A\n', I_DM);

% Gate source voltage plateau calculations
fprintf('7: Gate Plateau Voltage Calculations\n');

V_gs_pl = V_th + sqrt(2 * I_D_max / (mu_n * C_ox * (W/L)));
fprintf('Plateau voltage Vgs_pl = %.4f V\n\n', V_gs_pl);

% Capacitance calculations
fprintf('8: Gate Capacitances Calculations\n');

% saturation
C_gs_sat = 2*W*L*C_ox/3 + W*L_ov*C_ox;
C_gd_sat = W*L_ov*C_ox;

% triode
C_gs_tri = W*L*C_ox/2 + W*L_ov*C_ox;
C_gd_tri = W*L*C_ox/2 + W*L_ov*C_ox;

% cutoff
C_gs_cut = W*L_ov*C_ox;
C_gd_cut = W*L_ov*C_ox;

fprintf('Saturation: Cgs = %.4e F, Cgd = %.4e F\n', C_gs_sat, C_gd_sat);
fprintf('Triode: Cgs = %.4e F, Cgd = %.4e F\n', C_gs_tri, C_gd_tri);
fprintf('Cutoff: Cgs = %.4e F, Cgd = %.4e F\n\n', C_gs_cut, C_gd_cut);

% Switching times calculations
fprintf('9: Switching times calculations\n');

% Use saturation-region capacitances for switching transitions
C_gs_sw = C_gs_sat;
C_gd_sw = C_gd_sat;
C_total = C_gs_sw + C_gd_sw;
C_iss = C_total;

% Current Fall time (turn-off)
if V_th > 0
    t_if = R_g * C_total * log(V_gs_pl / V_th);
else
    % For depletion-mode: use |Vth| in denominator to avoid log of negative
    t_if = R_g * C_total * log(abs(V_gs_pl) / abs(V_th));
end

% Current Rise time (turn-on)
denom_tr = V_g - V_gs_pl;
if denom_tr <= 0
    warning('V_g - V_gs_pl <= 0');
    t_ir = NaN;
else
    t_ir = R_g * C_total * log((V_g - V_th) / (V_g - V_gs_pl));
end

% Turn on and off delay

```

```

L_s = 2e-9;
t_ind = L_s * I_D_max / (V_g - V_th);
td_on = R_g * C_total * log(V_g / (V_g - V_th)) + t_ind;
td_off = R_g * C_total * log(V_g / V_gs_pl) + t_ind;

% Voltage Fall Time
t_vf = R_g * C_gd_sw * V_ds / (V_g - V_gs_pl);

% Voltage Rise Time
t_vr = R_g * C_gd_sw * V_ds / V_gs_pl;

% Total Rise and Fall Times
t_r = t_ir + t_vf;
t_f = t_if + t_vr;

f_sw = 1 / (t_f + t_r + td_on + td_off);

% Gate charge calculations
Q_gs1 = C_iss*V_th;
Q_gs2 = C_iss*(V_gs_pl - V_th);
Q_gs = Q_gs1 + Q_gs2;

% Qgd related to voltage fall time
Q_gd = (V_g - V_gs_pl)*t_vf/R_g;

Q_g = Q_gs + Q_gd;

% Output Switching Times
fprintf('Fall time tf = %.4e s (%.4f ns)\n', t_f, t_f*1e9);
fprintf('Rise time tr = %.4e s (%.4f ns)\n', t_r, t_r*1e9);
fprintf('On Delay tdon = %.4e s (%.4f ns)\n', td_on, td_on*1e9);
fprintf('Off Delay toff = %.4e s (%.4f ns)\n', td_off, td_off*1e9);
fprintf('Switching frequency fsw = %.4e Hz (%.4f MHz)\n\n', f_sw, f_sw*1e-6);

% Output Gate Charge Parameters
fprintf('10: Gate Charge Parameters\n');
fprintf('Charge to Vth Qgs1 = %.4e C (%.4f nC)\n', Q_gs1, Q_gs1*1e9);
fprintf('Charge Vth to Plt. Qgs2 = %.4e C (%.4f nC)\n', Q_gs2, Q_gs2*1e9);
fprintf('Total Gate-Source Qgs = %.4e C (%.4f nC)\n', Q_gs, Q_gs*1e9);
fprintf('Miller Charge Qgd = %.4e C (%.4f nC)\n', Q_gd, Q_gd*1e9);
fprintf('Total Gate Charge Qg = %.4e C (%.4f nC)\n\n', Q_g, Q_g*1e9);
fprintf('Input Capacitance Ciss = %.4e F (%.4f nF)\n\n', C_iss, C_iss*1e9);

%p-n junction calculations
fprintf('11: p-n Junction Calculations\n');

% Built-in potential
phi_0 = V_t * log(Na * Nd / ni^2); % [V]
fprintf('Built-in potential phi_0 = %.4f V\n', phi_0);

% Junction area (based on device width and depletion extent)
A_j = W*x_dT;
fprintf('Junction area Aj = %.4e m2\n', A_j);

% Zero-bias junction capacitance per unit area

```

```

C_j0 = sqrt(e_s * q * (Na*Nd/(Na+Nd)) / (2*phi_0)); % [F/m2]
fprintf('Zero-bias junction cap Cj0 = %.4e F/m2\n', C_j0);

% Equivalent capacitance factor Keq for voltage swing
V1_junc = 0; % [V] (forward or zero bias)
V2_junc = -V_ds; % [V] (full reverse bias = -Vds)

K_eq = (-2*sqrt(phi_0)/(V2_junc - V1_junc)) * (sqrt(phi_0 - V2_junc) - sqrt(phi_0 - V1_junc));
fprintf('Keq (0 -> -%.0fV) Keq = %.6f\n\n', abs(V2_junc), K_eq);

% Find Cds, Coss
C_ds = A_j*C_j0*K_eq;
fprintf('Cds = %.4e F\n', C_ds);
C_oss = C_ds + C_gd_sat;
fprintf('C_oss = %.4e F\n\n', C_oss);

% dv/dt calculation
dvdt_est = (V_ds) / t_f;

fprintf('12: dv/dt Calculation\n');
fprintf('Estimated dv/dt = %.4e V/s (%.2f V/ns)\n\n', dvdt_est, dvdt_est/1e9);

% Power analysis calculations
fprintf('13: Power Analysis Calculations\n');

% Conduction power loss
% Assuming a 50% duty cycle for worst-case estimation
duty = 0.5;
P_cond = (I_D_max^2) * R_on * duty;
fprintf('Conduction Loss P_cond = %.4f W\n', P_cond);

% Coss power loss
E_coss = 0.5 * C_oss * V_ds^2; % [J]

fsw_p = 0;

f_range = linspace(100e3, 2e6, 20); % Sweep from 100 kHz to 2 MHz
P_total_sweep = zeros(size(f_range));

for i = 1:length(f_range)
    fs = f_range(i);
    P_total_sweep(i) = P_cond + ...
        (V_ds * I_D_max * ((t_r + t_f)/2) * fs) + ...
        (E_coss * fs) + ...
        (Q_g * V_g * fs);
end

P_max_thermal = (Tj_max - T_case) / Rth_jc;

if P_max_thermal < P_cond
    fsw_p = 0;
    warning('Conduction losses alone exceed the thermal limit!');
elseif max(P_total_sweep) >= P_max_thermal
    fsw_p = interp1(P_total_sweep, f_range, P_max_thermal);

```

```

    fprintf('Thermal Limit Frequency (fsw_p): %.2f kHz\n', fsw_p/1e3);
else
    fsw_p = max(f_range);
    warning('Thermal limit not reached within 2 MHz sweep.');
```

end

% Switching power loss

```

P_sw = V_ds * I_D_max * ((t_r + t_f) / 2) * fsw_p;
fprintf('Switching Loss P_sw = %.4f W\n', P_sw);
```

% Power loss: $P = E * f_{sw}$

```

P_coss = E_coss * fsw_p; % [W]
```

```

fprintf('Energy per cycle E_coss = %.4e J\n', E_coss);
fprintf('Coss Loss P_coss = %.4f W\n', P_coss);
```

% Gate drive power loss

```

Q_g = C_iss * V_g;
P_gate = Q_g * V_g * fsw_p;
fprintf('Gate Drive Loss P_gate = %.4f W\n', P_gate);
```

% Total power dissipation

```

P_total = P_cond + P_sw + P_coss + P_gate;
fprintf('TOTAL POWER LOSS P_tot = %.4f W\n', P_total);
```

% Quick Summary Table

```

fprintf('-----\n');
fprintf('Model Results\n');
fprintf('-----\n');
```

% fprintf('Thermal voltage	V_t = %.4f V\n', V_t);
% fprintf('GaAs bandgap	Eg = %.4f eV\n', Eg_T);
% fprintf('Intrinsic concentration	ni = %.3e m-3\n', ni);
fprintf('Channel mobility	mu_n = %.1f cm2/Vs\n', mu_n*1e4);
% fprintf('Fermi potential	phi_fp = %.4f V\n', phi_fp);
fprintf('Oxide capacitor density	Cox = %.4e F/m2\n', C_ox);
fprintf('Oxide Electric Field	E_ox = %.4f MV/cm\n', E_ox);
fprintf('Max depletion width	x_dT = %.4f mum\n', x_dT*1e6);
fprintf('Threshold voltage	Vth = %.4f V\n', V_th);
fprintf('Plateau voltage	Vgs_pl = %.4f V\n', V_gs_pl);
% fprintf('Drift doping	Nb = %.3e m-3\n', Nb);
fprintf('Drift width	W_drift = %.4f mum\n', W_drift*1e6);
fprintf('Drain current continuous	Id = %.4f A\n', I_D_max);
fprintf('On-resistance	R_on = %.4f ohm\n', R_on);
fprintf('Transit frequency	fT = %.4f GHz\n', f_T*1e-9);
fprintf('Rise time	tr = %.4f ns\n', t_r*1e9);
fprintf('Fall time	tf = %.4f ns\n', t_f*1e9);
fprintf('On Time	tdon = %.4f ns\n', td_on*1e9);
fprintf('Off Time	toff = %.4f ns\n', td_off*1e9);
fprintf('Switching freq. Max	f_sw = %.4f MHz\n', f_sw*1e-6);
fprintf('Switching freq. Op	f_sw = %.4f MHz\n', fsw_p*1e-6);
fprintf('Transconductance	gm = %.4f S (A/V)\n', gm);
% fprintf('Built-in potential	phi_0 = %.4f V\n', phi_0);
% fprintf('Junc. cap density	Cj0 = %.4e F/m2\n', C_j0);